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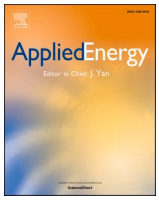

Targeted and cost-efficient solar cleaning: A novel RTVA approach to robotic path planning

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HIGHLIGHTS

- A smart robotic solar panel cleaning system integrating dual-axis motion and energy-aware control was developed.
- A lightweight image-processing pipeline enables selective cleaning of only detected soiled regions.
- The proposed Recursive Tabular Validation Algorithm (RTVA) optimizes cleaning routes with intersection-free, energy-efficient paths.
- RTVA achieves up to **19 % energy savings** and **faster runtime** compared to conventional algorithms such as NNA and 2-Opt.
- The framework demonstrates scalability for large photovoltaic fields and offers future potential for multi-agent or drone-based systems.

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ABSTRACT

The efficiency of photovoltaic systems can be significantly compromised by surface soiling, leading to substantial energy losses. While autonomous cleaning systems are increasingly deployed, most existing approaches rely on uniform surface cleaning or time-based scheduling, often ignoring the uneven distribution of contaminants. In this study, a contamination-aware robotic cleaning system is developed, integrating real-time image processing, environmental sensing, and energy-efficient path planning. The system identifies soiled regions through vision-based analysis and schedules cleaning actions according to environmental conditions. A novel routing algorithm, the Recursive Tabular Validation Algorithm, is introduced to solve the Traveling Salesman Problem in the context of partial and targeted cleaning. Unlike conventional heuristics that optimize the visit order, this algorithm prioritizes edge selection and employs a recursive table validation mechanism combined with path-intersection analysis to eliminate suboptimal routes. The prototype comprises four 25 W photovoltaic panels, and scalability simulations were conducted to evaluate algorithmic performance under larger problem sizes. Experimental results demonstrate a 19 % reduction in cleaning energy consumption, with the proposed algorithm outperforming the Nearest Neighbor and the Nearest Neighbor +2-Opt combinations by 17 % and 15 %, respectively. By prioritizing energy-aware route planning and condition-based cleaning, this study introduces a novel contribution to the field of autonomous PV maintenance systems and offers a scalable solution for enhancing the operational efficiency of solar farms.

1. Introduction

Solar photovoltaic (PV) systems have become an essential component of sustainable energy generation. However, surface soiling—caused by dust, pollution, and organic residues—can substantially degrade energy output [1], with reported efficiency losses ranging from 25 % to 35 % depending on environmental conditions [2]. In extreme cases, even light dust layers (e.g., 25 mg/cm^2) have been shown to reduce panel output nearly to zero [3]. In arid and semi-arid regions, where dust accumulation is more frequent, energy loss can become economically significant if not addressed with timely and efficient cleaning [4]. Beyond energy loss, dust accumulation also increases the surface temperature of PV modules, intensifies efficiency degradation due to cell overheating [5]. Prolonged soiling may further result in irreversible

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degradation of panel materials, accelerating system aging and increasing long-term operational costs [6].

In a three-month study conducted in Egypt, five solar panels were installed under identical conditions but cleaned at different intervals. The resulting power losses were recorded as 2.39 %, 10.89 %, 30.16 %, and 41.97 % for weekly, monthly, bi-monthly, and quarterly cleaning schedules, respectively [7]. Since the soiling rate varies geographically, cleaning intervals must also be adapted to local conditions. While the impact of periodic cleaning on system efficiency is widely discussed, several studies focus on calendar-based optimization to determine the most suitable cleaning intervals [8–10]. Singh et al. (2023) employed a biography-based Helianthus Optimization (HBBHO) algorithm in combination with Deep Learning (DL) tools to optimize cleaning schedules. Using Support Vector Machines (SVM) and Convolutional Neural Networks (CNN), they estimated dust accumulation levels from panel images [11]. Similarly, Zhang et al. (2025) applied advanced machine-learning (ML) algorithms—Variational Mode Decomposition (VMD) and Conditional Generative Adversarial Networks (cGANs)—to design an intelligent system that recommends optimal cleaning schedules for large-scale PV farms [12]. These data-driven approaches demonstrate the growing interest in adaptive cleaning interval optimization, though they primarily focus on scheduling rather than energy-efficient cleaning operations.

To mitigate soiling-related losses, various cleaning strategies have been proposed, ranging from manual periodic cleaning [7] to autonomous robotic systems. While manual washing [8,10] remains a common practice in small-scale installations due to its low initial cost, it poses significant limitations—especially for rooftop applications—where safety concerns and potential damage to panel surfaces make manual intervention impractical. Semi-automated cleaning systems represent an intermediate stage and typically operate on predefined schedules derived from empirical estimations, often without considering actual contamination levels. Portable semi-automated robots—including ultrasonic-guided, wind-resistant, and traversing designs [13–15]—have improved coverage and reduced manual labor but still require human intervention for positioning and inter-row transfer, reintroducing occupational risks. Most such robots assume uniform contamination, leading to excessive energy use and unnecessary mechanical wear. Fixed or panel-mounted systems such as belt-driven mechanisms [16] and waterless dual-stage robots [17] eliminate deployment requirements but lack mobility, adaptability, and sensing-based optimization. These limitations highlight the need for intelligent autonomous systems capable of selective and energy-efficient cleaning.

In contrast, fully autonomous systems incorporate sensor-based control strategies that enable adaptive cleaning decisions based on environmental and performance data. For instance, fuzzy-logic-based cleaning robots utilize power, temperature, contamination, and shading factors as inputs [18], while optical particle counters have been employed for surface contamination detection and automated activation of water-based cleaning mechanisms [19]. A recent study [20] proposed a waterless self-cleaning system using a transparent rolling film to remove dust and maintain surface transmittance, where contamination levels are detected through the Ares soiling station to trigger the cleaning cycle. Consequently, despite these advancements, such systems generally lack spatial awareness and energy optimization, highlighting the need for a more selective and intelligent robotic cleaning framework.

Moreover, image-processing-based frameworks have contributed significantly to advancing autonomous cleaning technologies. Previous research has explored aerial image analysis using pattern-matching algorithms to identify soiled panels and trigger automated cleaning commands [21]. Other studies developed motor-driven wiper mechanisms activated by vision-based contamination detection [22]. A robot prototype combining an IP camera and LDR sensors was also implemented, where real-time imagery and irradiance data were processed by an embedded controller [23]. More recently, compact single-panel cleaning robots integrating DC servo motors, dual brushes, and waterless mechanisms have achieved high classification accuracy of soiling patterns using DL models [24]. Despite these advancements, most existing systems still assume uniform contamination, leading to unnecessary energy use and redundant cleaning operations.

After reviewing previous research, a comparative summary of the most relevant studies is provided in Table 1, highlighting their key contributions and limitations to contextualize the novelty of the paper.

As summarized in Table 1, existing autonomous and semi-autonomous approaches exhibit fundamental trade-offs between accuracy, computational complexity, and energy efficiency. Moreover, recent research indicates that robotic cleaning platforms continue to face multidisciplinary challenges—from mechanical aspects such as stability, control, and power management to operational issues involving maintenance, communication, and skilled-labor requirements [25]. Limitations in modeling, particularly in radiation estimation, soiling loss indices, and conversion efficiency, further complicate the development of robust and scalable systems. These observations collectively underscore the need for a framework that not only optimizes energy consumption but also simplifies control logic, enhances autonomous decision-making capabilities, and provides stable robotics.

Previous research on fully autonomous PV cleaning has largely relied on two contrasting strategies: DL-based decision systems, which achieve accurate contamination detection but incur high computational and energy costs; and full-surface cleaning robots, which operate at fixed intervals regardless of actual soiling distribution. These approaches often neglect the trade-off between computational efficiency and cleaning energy consumption.

In response to this gap, the present study introduces a lightweight, contamination-aware robotic cleaning framework that integrates efficient image processing with sensor-based decision-making to perform targeted cleaning only on contaminated regions. At the core of this system lies the Recursive Tabular Validation Algorithm (RTVA)—a novel routing method that reformulates the Traveling Salesman Problem (TSP) from a node-ordering perspective (as in Nearest Neighbor or metaheuristic algorithms) into an edge-selection and recursive validation framework, substantially reducing redundant path traversal. Accelerating real-time cleaning decisions while minimizing operational costs is therefore a key consideration of this study. This integration of lightweight vision, and energy-optimized routing represents the primary innovation of the study.

The remainder of this paper is organized as follows: Section 2 presents the Methodology, detailing the overall system design, control architecture, energy modeling, image-processing pipeline, and the proposed RTVA for route optimization. Section 3 discusses the Results and Discussion, where the experimental findings and comparative analyses with existing methods are presented and interpreted. Finally, Section 4 concludes the study with key Conclusions and Recommendations, including identified limitations and directions for future research.

2. Smart solar panel cleaning system

The proposed system establishes an adaptive and energy-aware decision framework that continuously evaluates whether cleaning is economically and energetically justified. In the control logic, the expected future energy loss—predicted from real-time power measurements and historical performance data—is dynamically compared with the cleaning energy consumption required to remove the detected contaminants. This comparison forms a continuously updated, closed-loop decision layer that enables real-time optimization of cleaning operations under varying soiling and environmental conditions. The RTVA functions as the driving mechanism of this decision process, determining the most energy-efficient route among candidate cleaning paths while avoiding redundant traversals.

To achieve these objectives, the system integrates three synergistic modules.

First, a mechanical cleaning platform was developed that can reach

Table 1
Comparative summary of existing PV panel cleaning strategies.

Study	Approach / Technique	Key Contribution	Limitation	Relevance to This Work
Manual Cleaning [7,8,10]	Manual water or brush cleaning	Low initial cost, simple operation	Unsafe for rooftops; labor-intensive; risk of panel damage	Motivates need for automation
Portable Semi-Automated Systems [13–15]	Wind-resistant, ultrasonic-guided, or traversing robots	Improved surface coverage and reduced manual labor	Require human deployment; assume uniform soiling	Supports need for fixed-systems and automation
Fixed or Panel-Mounted Semi-Automated Systems [16,17]	Belt-driven or dual-stage (air-blower + brush) waterless robots	Permanent installation reduces manual handling	No intelligent control or energy optimization	Highlights need for adaptive stationary systems
Fully Autonomous Systems [18–20]	Fuzzy-logic robot, optical dust-sensor and Ares soiling station	Automatic activation based on real-time conditions	Treat entire panel as uniformly dirty; no spatial awareness or energy optimization	Reveals gap addressed by RTVA on localized cleaning and routing
Image-Processing and Computer-Vision Systems [21–24]	Visual dirt detection, image classification	Enabled real-time dust detection and cleaning trigger	High computational load; limited energy analysis	Lays foundation for lightweight vision-guided RTVA framework and spatial dirt detection
This Work (RTVA)	Image-based contamination detection + Recursive Tabular Validation Algorithm	Lightweight, spatial contamination-sensitive, energy-optimized route planning	Validated on small-scale prototype; extreme conditions not yet tested	Fills identified gap and provides scalable foundation for autonomous PV maintenance

every critical point of the panel surface with precise actuation, enabling fine motion control and efficient contact-pressure regulation, with the routing process available to optimize movement efficiency and minimize unnecessary travel across the panel.

Second, a multi-sensor contamination-detection architecture combines image processing with environmental sensing to detect, validate, and quantify the degree of soiling.

Third, the image-processing pipeline serves as an intelligent interface between the sensing system and the RTVA algorithm: contaminated regions are identified, segmented, and grouped into coordinate clusters that define candidate cleaning targets. These clustered coordinates are then used by RTVA to iteratively construct and validate the most energy-efficient trajectory that connects all detected contaminated points.

The overall methodology — including data acquisition, decision logic, and route optimization—is governed by several modeling assumptions that support the RTVA framework and maintain the focus of this study on algorithmic design. These assumptions, explained in detail in Section 2.1.2, define the simplified operational conditions used to evaluate cleaning efficiency, energy-loss estimation, and control thresholds. They ensure conceptual clarity by abstracting the complex interactions between mechanical efficiency, environmental variability, and energy-loss prediction.

2.1. System overview and control architecture

A fully autonomous robotic cleaning prototype was developed, specifically designed for rooftop PV installations. The system features a dual-axis motion structure mounted directly on the PV panel frame. Unlike portable semi-automated systems [13–15], which require manual positioning or inter-row transfers, and panel-mounted systems [17,18,22] limited to single-axis operation, the proposed design integrates both horizontal and vertical motion axes on a single structural frame. This configuration eliminates human interaction during operation and minimizes both structural damage to the panels and safety risks, while removing the need for external deployment mechanisms or rails, thereby reducing mechanical complexity and improving stability under rooftop conditions. The overall configuration and subsystem interactions are illustrated in Fig. 1.

As illustrated in Fig. 1, the robotic platform integrates both mechanical and control subsystems. The mechanical assembly consists of an X–Y motion structure driven by stepper motors through ball-screw mechanisms that provide high-precision positioning. Three stepper motors are employed: one for horizontal (X-axis) translation of the brush and two for vertical (Y-axis) motion to ensure balanced carriage movement, which directs localized, contamination-aware cleaning. Both

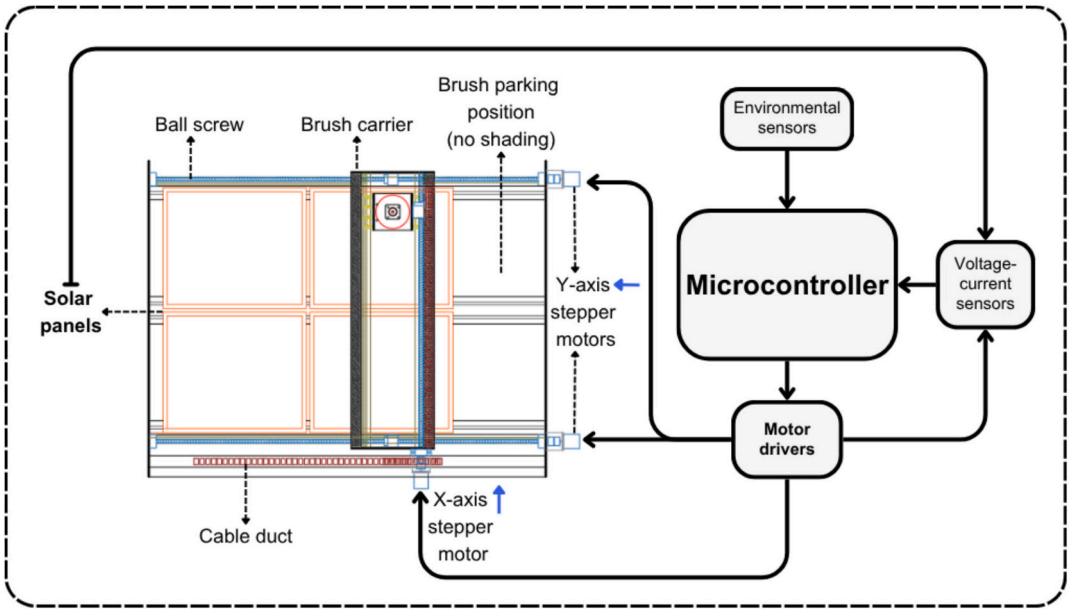

Fig. 1. Schematic representation of the dual-axis autonomous robotic cleaning system and its control interactions.

axes utilize ball screw mechanisms for high-precision linear positioning, while shaft couplings, flanged nuts, and linear bearings are implemented to maintain mechanical stability and uniform contact pressure. A dedicated parking position, located outside the active panel area, prevents shading when the system is idle. The control subsystem includes a microcontroller, motor-driver unit, and multiple sensor inputs. Sensors such as irradiance and temperature monitor environmental conditions, while voltage–current sensors record both PV generation and cleaning energy consumption. The microcontroller synchronizes these inputs and generates step-signal pulses for coordinated motion control along the X and Y axes.

The constructed prototype, shown in Fig. 2, was deployed on a small-scale PV array comprising four 25 W modules installed at the Solar and Wind Energy Research Center of Afyon Kocatepe University. The lead-screw-driven motion along both axes provided smooth and accurate bidirectional movement of the cleaning brush. To ensure reliable validation of the contamination-detection and power-loss estimation process, a reference panel was installed adjacent to the test array with the same tilt and orientation. This panel remained uncleaned during experiments and served as a benchmark for evaluating the performance of the image-processing pipeline and the RTVA-based decision framework under varying soiling levels.

Controlled soiling conditions were established by applying fine dust on the test modules, allowing repeatable assessment of the routing performance. This mechanical-control configuration ensures precise positioning, balanced torque distribution. Together, the hardware design and embedded control logic provide a robust experimental platform for evaluating the energy-aware routing performance of the RTVA algorithm described in the subsequent sections.

2.1.1. Control architecture and hardware implementation

The overall control system architecture is illustrated in Fig. 3, which consists of both the physical control unit and its schematic interaction layout.

In Fig. 3a, the real hardware setup is presented, including the sealed microcontroller enclosure, motor drivers, power-supply module, and sensor interfaces. The system is powered by a 24 V DC line derived from the grid through an AC/DC converter. The control unit is built around an Arduino Mega 2560 microcontroller featuring an ATmega2560 processor operating at 16 MHz, with 54 digital I/O pins (15 PWM outputs), 16 analog inputs (10-bit ADC), and 256 kB of flash memory supported by 8 kB SRAM and 4 kB EEPROM, enabling parallel sensor monitoring and synchronized motor control.

The system integrates multiple sensors for real-time environmental and operational monitoring: a temperature and humidity sensor (DHT-22) for ambient conditions, current sensors (ACS724) for measuring both PV output and cleaning power consumption, and a simple Arduino rain sensor to prevent cleaning during precipitation and to support the estimation of natural cleaning periods following rainfall events.

In Fig. 3b, a schematic representation of the data and control flow is presented. A fixed camera positioned above the PV array captures real-time panel images, which are transmitted to a PC for processing via a Python–OpenCV framework. The resulting contamination map and coordinate data are then sent to the microcontroller, which evaluates the necessity of cleaning through the decision logic illustrated in Fig. 4.

In this stage, the RTVA is embedded as the primary optimization layer within the decision logic, determining the most energy-efficient route based on contamination position data.

2.1.2. Analytical framework and simplifying assumptions for RTVA validation

This article has been prepared to introduce the RTVA as a supporting component for the real-time decision-making mechanism (Fig. 4) developed during the design of an energy-aware intelligent solar panel cleaning robot. Therefore, it should be evaluated within this context.

In the decision-making architecture, the future energy loss is estimated based on irradiance and temperature in combination with the measured instantaneous panel efficiency. The instantaneous efficiency or soiling rate (SR) is determined by comparing the measured power with the expected output derived from the measured environmental data, which represents the power of a clean panel under the same conditions. Recent studies [26] have shown that the instantaneous SR can also be predicted using ML models trained on historical environmental parameters. The total future energy loss is calculated as:

$$FL = \int_{0}^{CT} [\, P_{CSP}(G(t)\,, T(t)\,) \cdot S(D,t) - P_{USP}(G(t)\,, T(t)\,) \cdot S(D,t)\ ] \cdot \delta t \qquad (1)$$

where P_{CSP} denotes the cleaned system power calculated under predicted irradiance $G(t)$ and ambient temperature $T(t)$, while P_{USP} represents the uncleaned system power estimated using an efficiency factor relative to the cleaned state. CT denotes the total cleaning period, during which rainfall events are treated as natural cleaning cycles that limit the algorithm's operational window. Predefined periodic cleaning intervals can also be analyzed separately to compare periodic and real-time decision frameworks.

S represents the soiling-speed function for a specific site, which varies depending on environmental and geographic conditions and typically decreases exponentially over time. Here, D represents the

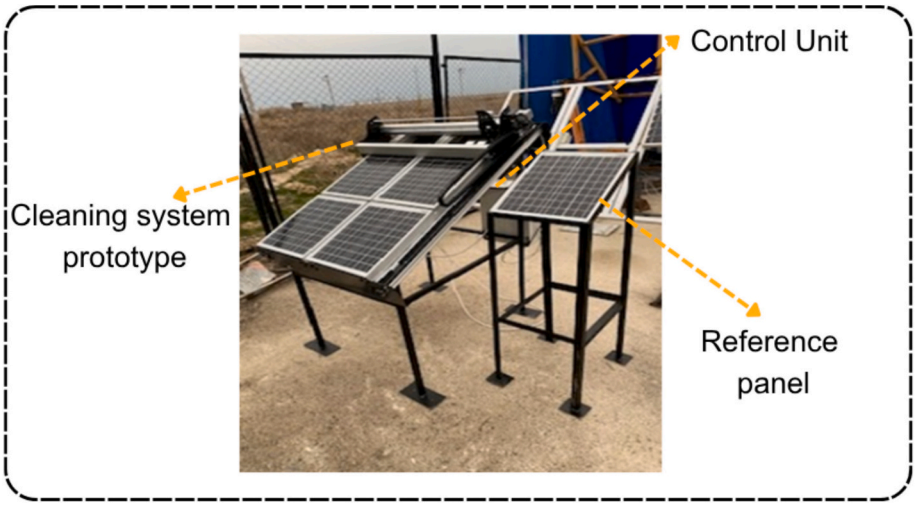

Fig. 2. Photograph of the constructed prototype installed on the rooftop PV testbed.

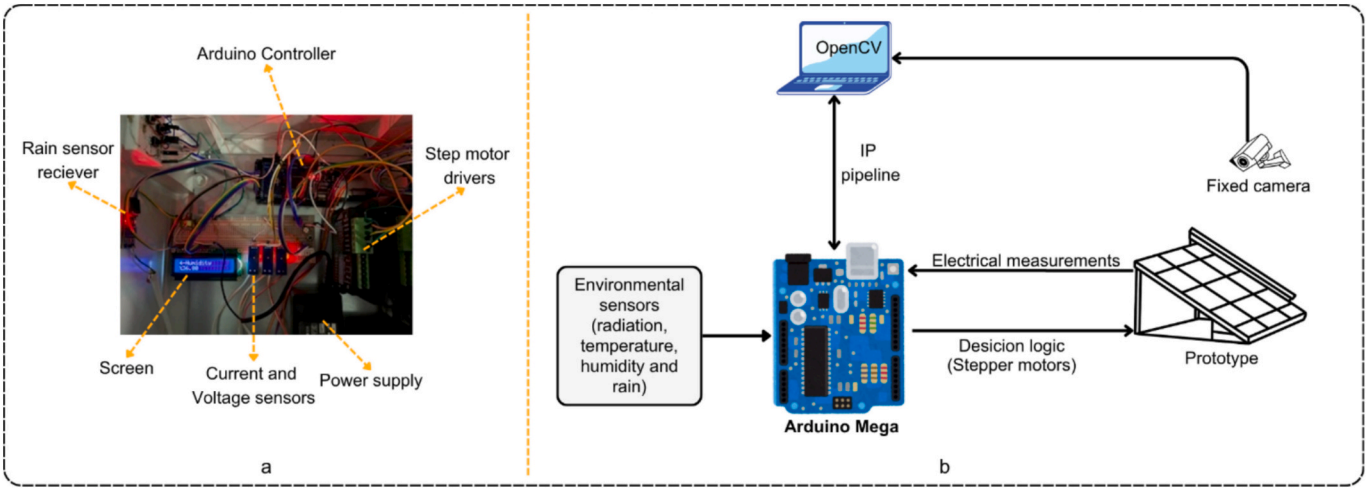

Fig. 3. Hardware and control architecture of the developed cleaning system: (a) experimental setup; (b) data acquisition and control flow.

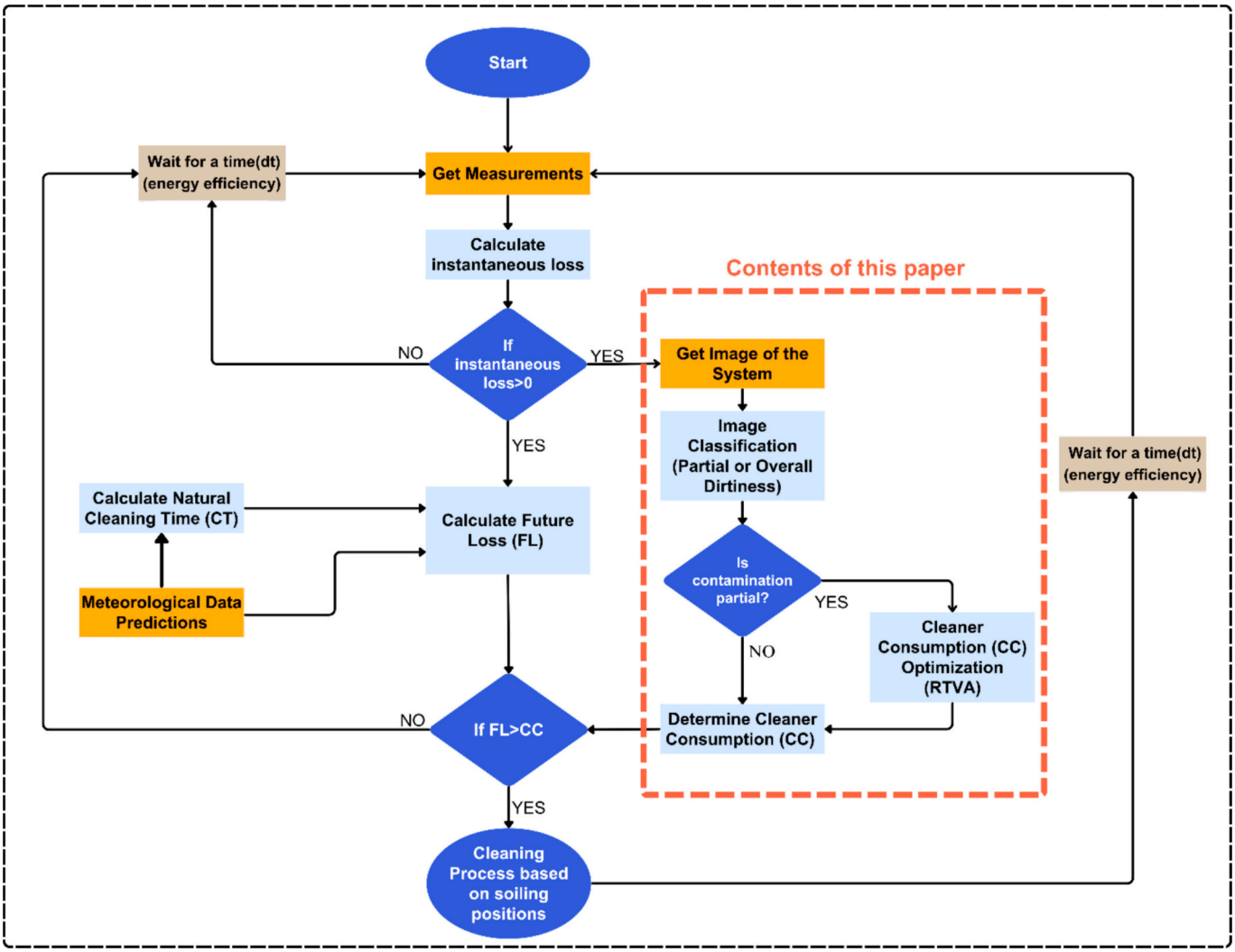

Fig. 4. Decision-making flowchart of contamination detection and cleaning activation logic implemented on the microcontroller. The red dashed box highlights the section developed in this study (contents of this paper). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

initial dirtiness level, which defines the starting point (t) of the $S(D, t)$ and significantly affects its overall behavior.

Meteorological parameters such as solar irradiance and temperature have been extensively modeled in the literature. Although instantaneous prediction accuracy may be limited due to the high variability of weather conditions, models with low Mean Bias Error (MBE) [27] over a defined time interval (CT) still allow sufficiently accurate estimation of the overall energy loss.

The cleaned (P_{CSP}) and uncleaned (P_{USP}) system power outputs can be predicted from irradiance (G) and temperature (T):

$$P_m = P_{stc} \times \frac{G}{G_{stc}} \tag{2}$$

$$P_m{}' = P_m \times \left(1 - \mu_p \left(T_{p-} T_{stc}\right)\right) \tag{3}$$

where P_m and P_m' are the irradiance- and temperature-corrected system power outputs, respectively, both referenced to standard test condition (STC) values. T_p and T_{stc} are the panel and STC temperature, G_{stc} is the irradiance under STC, and μ_p denotes the temperature coefficient of power.

The panel temperature is the most influential factor affecting the output power and can be approximated from the ambient temperature as:

$$T_p = T + (T_{NOM} - 20) \times 1.25 \times \frac{G}{G_{stc}} \tag{4}$$

where T is the ambient temperature, T_{NOM} is the nominal operating cell temperature. Humidity data are also incorporated into the loss estimation process, since humidity indirectly affects panel temperature through convective cooling. Gholami et al. (2023) improved this temperature-correlation model by incorporating wind velocity to enhance the prediction accuracy of PV surface temperature [28]. Moreover, a hysteresis behavior in power output throughout the day has been observed in relation to temperature effects, attributed to varying irradiance gradients [29]. These correlation-based enhancements can improve overall prediction accuracy.

However, both the detailed mathematical formulations of these correlations and comprehensive uncertainty analyses in energy-loss estimation are considered beyond the scope of this study. Similarly, while various mechanical, electromechanical, and electrostatic cleaning methods have been discussed in recent studies [30], the cleaning efficiency of the proposed robot is assumed to remain high and consistent throughout the experiments. In addition, the mechanical energy model assumes constant drive efficiency and neglects transient effects such as motor inertia, acceleration, and deceleration phases. These simplifications were adopted to maintain analytical tractability during algorithm validation and to focus on the relative energy consumption between alternative routes rather than absolute power transients.

These clarifications ensure that the proposed framework is interpreted as a modular and extensible architecture, within which the RTVA algorithm can be seamlessly integrated into more complex PV monitoring and control systems without loss of generality. To support this integration, the energy consumption of the motion system must be analytically modeled, providing the cost basis for route optimization.

2.1.3. Energy consumption model

To quantify the energy consumed by the motion system, the number of motor steps required to traverse the panel surface is calculated as follows:

$$S_{x,all} = \left[\frac{H_x - H_b}{r}\right] \times ceil\left[\frac{H_y}{H_b}\right] \times \left[\frac{2}{\delta}\right] \tag{5}$$

$$S_{y,all} = 2 \times \left[\frac{H_y}{r}\right] \times \left[\frac{2}{\delta}\right] \tag{6}$$

Where H_x and H_y are the panel dimensions (cm), H_b is brush radius, r is the lead screw pitch, and δ is the motor step angle.

The measured parameters of the stepper motors and drivers used in the prototype are summarized in Table 2, where V_n denotes the supply voltage of the drivers, and t_s represents the step interval set by the microcontroller. The design employs a single motor on the X-axis and dual motors on the Y-axis, with currents drawn by the axes denoted as Icc_x and Icc_y, respectively. To maximize efficiency during vertical movement along the Y-axis, the mechanism was designed with a 30° inclination, resulting in two current values corresponding to upward and downward motion. Using the measured current data, the instantaneous power consumption of each motor driver operating at a 1 ms step interval is expressed as:

$$P_{cc} = V_n \times I_{cc} \tag{7}$$

The total energy required for the brush to traverse the full panel area (see Fig. 5) is given by:

$$E_{cc,all} = \left(S_{x,all} Pcc_x + S_{y,all} Pcc_y\right) \times t_s \tag{8}$$

Eq. 8 calculates the total energy consumption required for cleaning the entire surface, as illustrated in Fig. 5, whereas Eq. (9) computes the energy required to move between two contaminated points.

$$E_{cc} = \left[\frac{|X_1 - X_2|}{r} Pcc_x + \frac{|Y_1 - Y_2|}{r} Pcc_y\right] \times \left[\frac{2}{\delta}\right] \times t_s \tag{9}$$

This equation forms the fitness function used in the RTVA algorithm for route planning. It calculates the energy required for the cleaning brush to move from one contaminated region to another, allowing for cost-based prioritization of cleaning routes.

2.2. Lightweight image processing pipeline

Localized contamination—particularly bird droppings and dust clusters—can cause significant performance degradation in solar panels, even when only a small portion of the surface is affected. Therefore, detecting these contaminations with high spatial accuracy is critical for targeted cleaning and energy efficiency. Traditional image-processing (IP) approaches have typically focused on binary panel classification (i.e., clean vs. dirty), often employing ML or DL architectures. For example, CNNs [31], SVMs [32], and transfer-learning models such as SolNet—which outperforms ResNet50 and AlexNet in both speed and accuracy—have achieved high detection rates [33]. Dust pattern recognition has also been implemented using Elman deep networks [34] and transfer-learning architectures such as ResNet-18, VGG-16, and MobileNetV2 [35]. While these DL-based methods offer strong classification accuracy, they require large, labeled datasets, GPUs for inference, and are difficult to integrate into low-power embedded systems [36].

Alternative approaches using thermal [37], long-wave infrared (LWIR) imaging [38,39], and transmittance-based dust quantification [40], and HSV-based linear classifiers [41] to locate dirt and damage on PV modules based on pixel values.

In contrast, this study implements a lightweight image-processing pipeline capable of:

- Detecting partial soiling under variable illumination,

Table 2
Technical specifications and power consumption parameters of the stepper motors and drivers.

Parameter	Value
δ	1.8° ± %5
Icc_x	0.37 A.
Icc_y	↓1.07-↑1.10 A.
V_n	24 V.
t_s	1 ms.

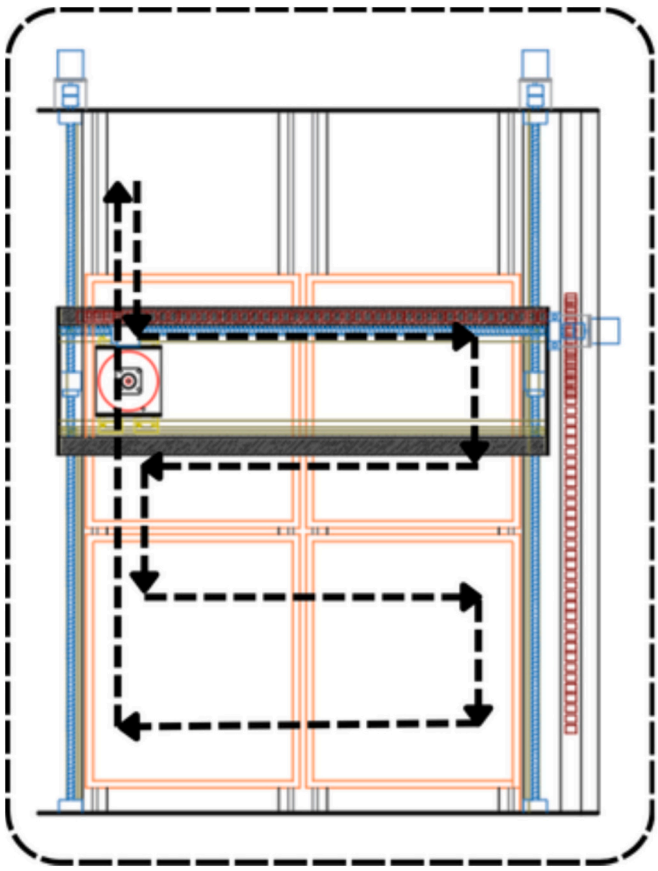

Fig. 5. Full-surface cleaning route of the prototype used for total energy consumption estimation.

- Locating contaminated regions with centroid coordinates and area sizes, (segmentation)
- Preparing input for an energy-aware route planning algorithm (RTVA).

The image processing workflow is presented in Fig. 6 and described in five sequential stages.

2.2.1. Panel boundary detection

The first step is identifying the boundaries of the panel in the image. The conversion of the RGB panel image to grayscale enables efficient processing within a one-dimensional intensity space, allowing the identification of regions with sharp intensity transitions that typically correspond to edges or contamination boundaries. Through adaptive thresholding, illumination variations across these regions are effectively compensated, ensuring consistent detection performance under diverse lighting conditions.

Subsequently, contour detection functions are applied to extract potential panel boundaries. However, some of the detected contours may not precisely correspond to the actual panel edges. To refine the results, the Douglas–Peucker (DP) algorithm [42] is employed with adaptive epsilon (tolerance) values to simplify contour geometry and accurately identify the four dominant panel corners (Fig. 7).

In Fig. 7(b), it can be observed that bright reflections on the panel surface—particularly near the frame edges—are partially included within the largest contour due to the adaptive thresholding process. However, these small reflective regions do not affect the final boundary extraction, as the adaptive DP algorithm effectively compensates for such minor deviations during contour simplification.

2.2.2. Background masking

Once the panel boundaries are identified, a binary mask is generated using the four dominant corners coordinates to exclude background elements such as the sky, frame, and surrounding terrain (as shown in Fig. 8). This ensures that all further processing focuses exclusively on the panel surface.

2.2.3. Perspective correction (homography-warping)

To facilitate accurate area measurement and coordinate-based analysis, the tilted panel surface in the image is transformed to a rectified top-down view using homography. The resulting warped image aligns with the real panel dimensions in a 3620 × 4330 pixels frame, maintaining spatial proportionality, as illustrated in Fig. 9.

Although the quality of this transformation depends on the contour boundaries extracted through adaptive thresholding, the algorithm consistently detects a region encompassing both the cell area and a small portion of the panel frame. Consequently, the calculated center coordinates of the soiling regions remain unaffected by these minor geometric deviations.

2.2.4. Dirt detection

The rectified image is converted into HSV color space, which provides enhanced sensitivity to color changes associated with soiling. A predefined Hue filter (Fig. 10) is applied to isolate color regions corresponding to common soiling characteristics. Local irregularities in hue, saturation, and brightness are compared to the global average to detect anomalies.

The HSV representation is inherently more robust to illumination changes, since small fluctuations in external light intensity lead to minimal variations in hue compared with RGB [41]. Moreover, the HSV representation is preferred over RGB because it separates chromatic information (hue and saturation) from illumination (value), providing greater stability under varying lighting conditions.

The hue, saturation, and value ranges were determined empirically and optimized for the specific dust composition used in this study, based on evaluations under varying illumination conditions (morning, noon, and afternoon). Low saturation values were consistently observed along panel edges, cell interconnect lines, and specular sun reflections. Clean panel cell regions also exhibit low saturation and very low brightness values. Therefore, pixels with simultaneously low saturation and brightness were excluded from the filter to minimize false detections.

As illustrated in Fig. 11, when the panel boundaries are not tightly constrained—as discussed in previous sections—the HSV filter may partially include background areas near the frame. However, this effect remains negligible within the active panel region.

The final HSV thresholds were determined adaptively by calculating the mean ± 0.8 × standard deviation of the dusty regions identified in the image. The most frequently observed ranges are reported to demonstrate the principle of the lightweight HSV filtering approach. Using these thresholds, the detected contamination regions remained consistent across multiple tests, exhibiting less than 5 % deviation in contour area—confirming the robustness of the method under practical illumination variations.

Although detailed robustness testing with different contaminant types and panel designs (featuring varying cell-region colors) was not conducted in the present study, the theoretical advantage of the HSV representation—supported by prior empirical findings—demonstrates its suitability for lightweight, real-time implementations. The results further indicate that HSV-based detection provides sufficient accuracy for identifying contamination points on the small-scale prototype used in this study, ensuring reliable routing performance.

2.2.5. Feature extraction, clustering and route generation

The centroid coordinates and areas of the detected soiling regions are extracted and supplied as inputs to the RTVA routing algorithm. To prevent redundant or overlapping cleaning movements, a density-based

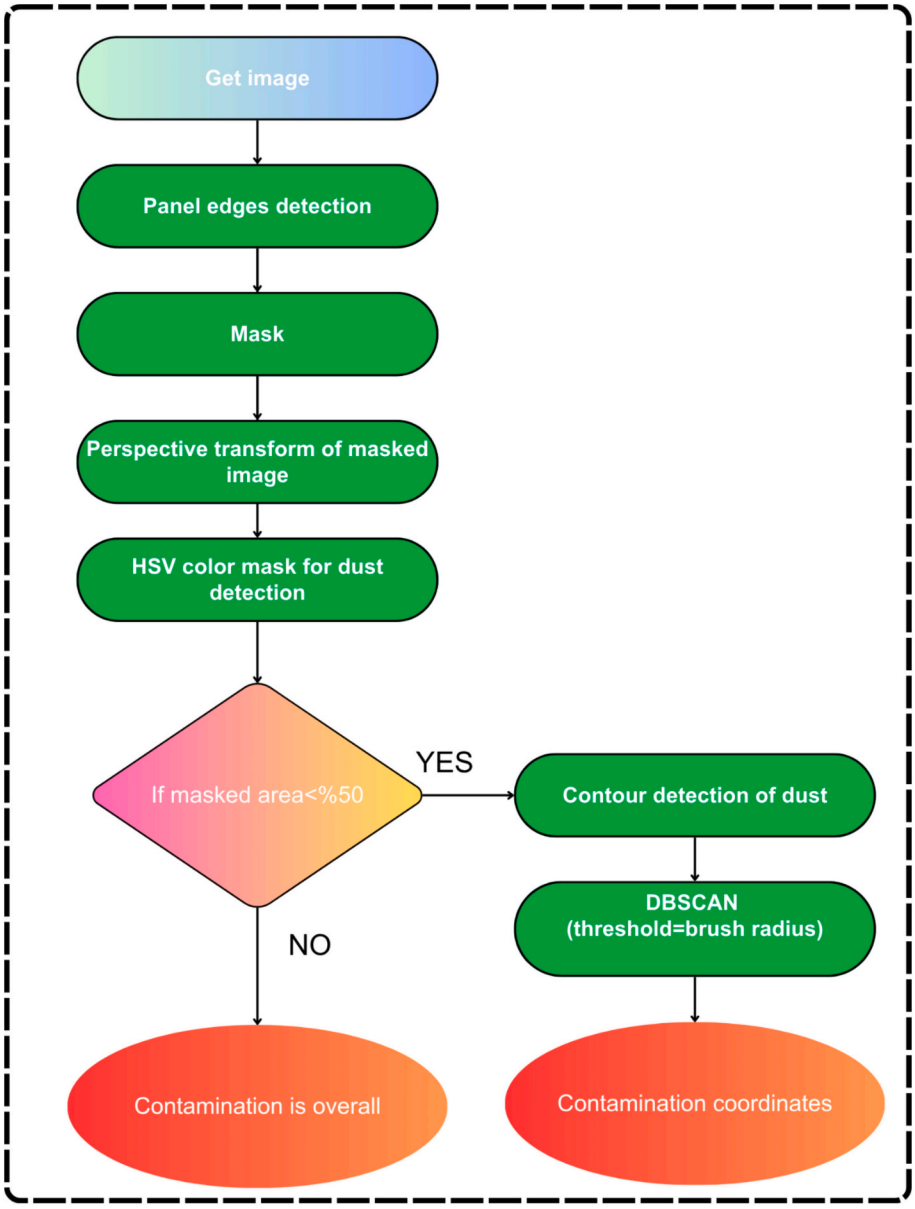

Fig. 6. Image-processing pipeline for contamination detection and coordinate extraction used in the proposed cleaning system.

spatial clustering (DBSCAN) algorithm [43] is applied to group contamination spots.

The DBSCAN hyperparameters were set as follows:

Epsilon (ε): 500 pixels. This distance threshold corresponds to approximately 1.2× the effective brush radius in the rectified image scale. The selected ε value defines the maximum spatial separation at which neighboring contamination regions are treated as belonging to the same cluster. In practical terms, this ensures that all soiling spots that can be cleaned within a single brush sweep are grouped together, avoiding unnecessary segmentation of contiguous dirt regions. The parameter was empirically tuned to achieve a balance between merging neighboring stains and maintaining sufficient spatial distinction among separate contamination clusters.

Minimum points (minPts): 1. A low minPts value was deliberately selected because contour extraction was preceded by an area-based filtering stage in which regions smaller than 30 pixels were removed. These small regions were considered either as noise or as negligible residues that are not meaningful for cleaning operations. Subsequently, a morphological refinement using a 15 × 15 kernel (erosion–dilation) was applied, which ensured that the remaining contours were stable, coherent, and well defined. Since the ε parameter already served to group neighboring contaminated regions within the effective brush radius, using higher minPts values after these preprocessing steps would have caused misclassification of valid contours as noise or undesired merging of separate contamination clusters.

Each resulting cluster is treated as an individual cleaning target, and the RTVA algorithm computes an energy-efficient route between the centroids of these clusters, minimizing redundant movements and optimizing total cleaning energy consumption.

2.3. *Proposed novel TSP algorithm (recursive tabular validation algorithm-RTVA)*

After identifying the coordinates of contaminated regions on the panel surface, an energy-efficient route must be generated for the cleaning brush to visit all these points. This requirement is modeled as a variation of the TSP—a classical NP-hard problem where the objective is to determine the shortest possible path that visits each node exactly once

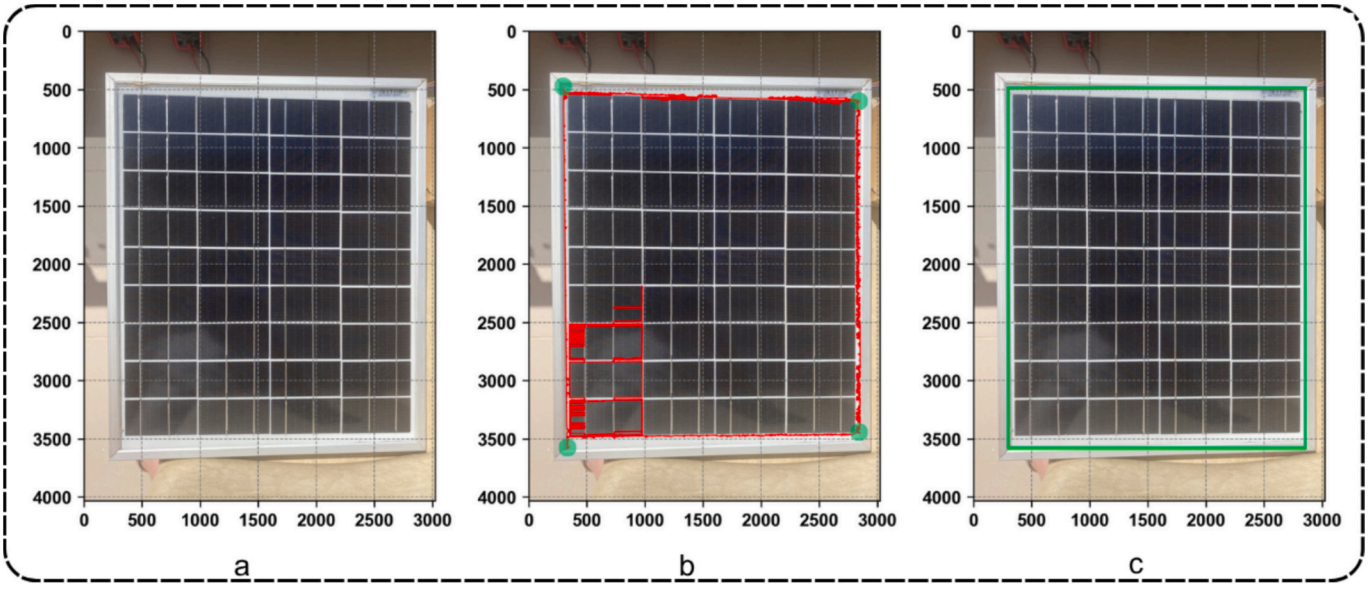

Fig. 7. Panel boundary detection process: (a) original image, (b) extracted largest contour with four dominant corners (green circles), and (c) final panel detection represented by a green rectangle derived from contour corners. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

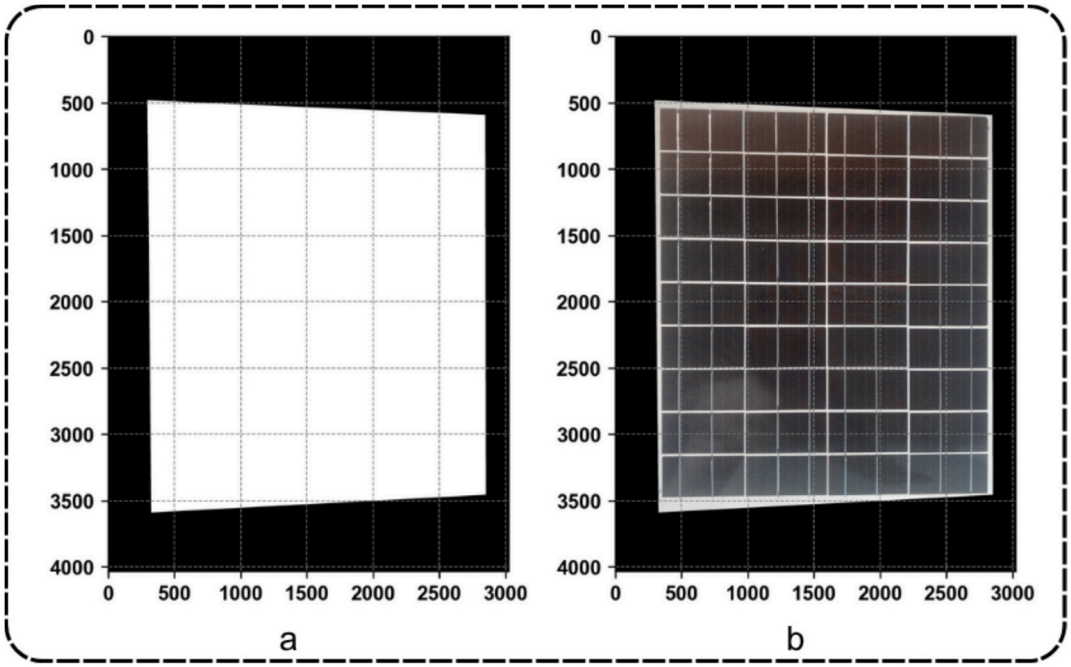

Fig. 8. Background masking process: (a) binary mask generated from panel corners, and (b) resulting masked image showing only the panel surface.

and returns to the starting point [44].

TSP and its variants have been widely applied in fields such as logistics, vehicle routing, manufacturing, maintenance scheduling, and disaster response [45,46]. For example, Jong and Ahmed (2024) used a fuzzy Binary Integer Programming (BIP) algorithm to optimize power line routing [47]. Numerous solution strategies have been developed, including mathematical programming, metaheuristic, ML, and DL approaches. Among these, population-based evolutionary algorithms—notably those derived from genetic algorithms (GA) [48]—are widely used for large-scale routing problems, though they typically involve slower computation times. Commonly used metaheuristic algorithms in the literature include GA [49], Discrete Particle Swarm Optimization (PSO) [50], Discrete Dragonfly Algorithm (DA) [51], Ant Colony Optimization (ACO) [52], Simulated Annealing (SA) [53], Tabu Search [54], and Bee Colony Optimization (BCO) [55]. In addition, Chen et al. (2017) utilized Glowworm Optimization (GO) by combining metaheuristics and classical, accelerating the optimization with 2-Opt algorithm [56]. While many of these methods are capable of finding near-optimal solutions, they often suffer from high computational complexity or converge issues, particularly in real-time or embedded applications.

To address these limitations, we propose a novel two-stage route optimization algorithm —called the Recursive Tabular Validation Algorithm. This algorithm is specifically designed for energy-aware path

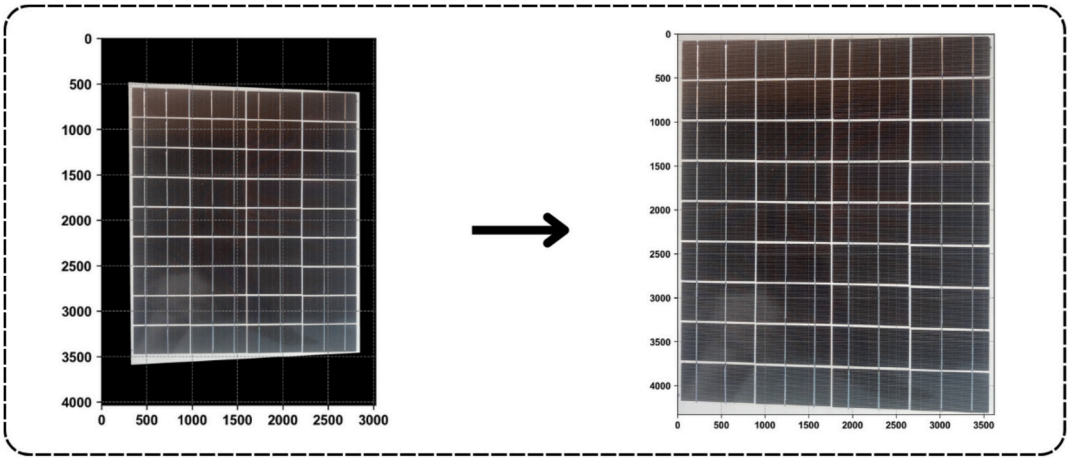

Fig. 9. Perspective transform to corresponding dimension (warped image 3620 × 4330 pixels).

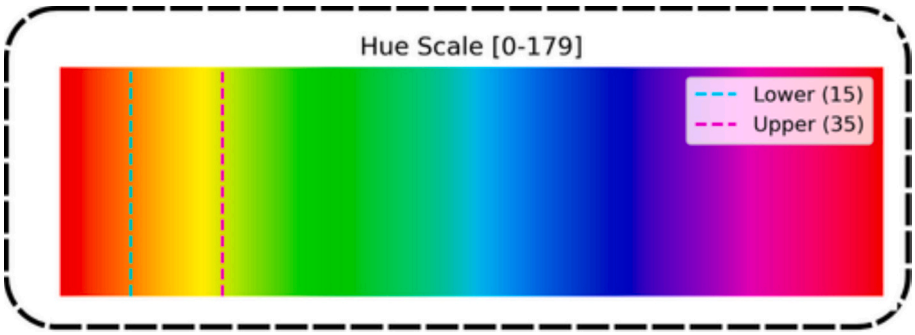

Fig. 10. Typical hue range corresponding to dust color values.

planning that starts like a greedy algorithm based on the selection of minimums and achieves the goals by validation operations.

Key Innovations of RTVA:

Path-Centric Optimization:

Unlike conventional TSP solutions that focus on visiting node sequences, RTVA evaluates and eliminates suboptimal inter-node paths regardless of visitation order. This reduces search space and improves convergence.

Two-Stage Architecture:

Stage 1 – Recursive Table Validation:

A rule-based recursive logic populates a cost table by evaluating all possible transitions between points, removing those that violate energy-efficiency constraints or geometrical feasibility.

Stage 2 – Path Intersection Optimizer:

This component checks for intersecting paths in the route and resolves them through a swapping heuristic, ensuring that the final path is both shortest and smoothest.

To provide a clear overview of the proposed architecture, Fig. 12 illustrates the RTVA framework as a block diagram.

2.3.1. Problem formulation

The optimization problem addressed in this study is a variation of the classical TSP, defined as follows:

Given a set of contaminated points on a 2D plane, determine the shortest path that visits each point exactly once and returns to the starting location, minimizing the total energy consumption of the cleaning robot.

Let:

- N = $\{V_1, V_2, \ldots, V_n\}$be the set of contaminated regions (nodes),
- D(V_i, V_j) represent the distance (or cost in energy) between nodes V_i and V_j.
- The objective is to find a permutation π of the nodes such that the total cost:

$$C(\pi) = \sum_{k=1}^{n-1} \mathrm{D}\left(V_{\pi_k}, V_{\pi_{k+1}}\right) + \mathrm{D}\left(V_{\pi_n}, V_{\pi_1}\right) \tag{10}$$

is minimized.

Traditional approaches aim to find an optimal node sequence π. In contrast, **RTVA focuses on path (edge) selection** using a recursive table validation process. This novel formulation not only reduces total travel distance but also minimizes the total energy consumption and cleaning time. It is particularly suited for embedded systems with limited computational resources, offering fast execution without compromising optimality.

2.3.2. Rule-based goal-oriented solution validation

The first stage of the proposed RTVA is a rule-based goal-oriented filtering mechanism that systematically validates and refines potential path selections. Unlike conventional TSP algorithms that optimize based on the visitation order of nodes, the proposed method represents the problem as the greedy selection of $\boldsymbol{N} - \boldsymbol{1}$ feasible paths between contaminated regions, where $\boldsymbol{N}$ is the number of nodes. This approach simplifies the solution representation and enables early elimination of suboptimal or infeasible routes.

2.3.2.1. Lookup table construction and validation logic. A lookup table (Fig. 13) is constructed from the initial distance (or energy cost) matrix. Each cell in the table represents a potential directed path between two nodes. Selected paths are color-coded based on compliance with the algorithm's routing rules:

- Red: Invalid paths that connect more than two other paths (i.e., cause node degree >2),
- Yellow: Semi-invalid paths that violate the rule from a single node's perspective.

A vector V is used to track the degree of each node. The algorithm recursively examines the table and eliminates red-marked paths first, as they represent hard violations. If no red paths remain, yellow paths are removed next to resolve soft violations. Blue-marked paths indicates that the selection is appropriate for that stage. The main goal is to ensure that all nodes have a maximum degree of 2 as illustrated in Fig. 11. In addition, to construct a complete TSP tour (sub-goal-1), closed loops are opened through a tournament elimination process and subsequently grouped into open loops. This hierarchical path elimination ensures topological correctness and accelerates convergence toward a valid tour.

This approach significantly reduces the solution space, focusing the search only on path combinations that meet routing constraints. Such early-stage pruning enhances both computational efficiency and the likelihood of discovering near-optimal paths with minimal energy cost. Various methods have been proposed to reduce the complexity of combinatorial optimization problems. For instance, the Reduce-Optimize-Expand (ROE) method incorporates fuzzy logic to shrink the search space dynamically [57]. Modifications of ACO have also been introduced to promote search space diversification [58]. In a similar vein, the proposed RTVA framework eliminates infeasible paths before optimization, enabling faster convergence without compromising solution quality.

A pseudocode representation of this validation stage is given in Pseudocode 1.

Pseudocode 1. Recursive Tabular Validation Algorithm

Pseudocode-1: Recursive Tabular Validation Algorithm

Input: *coordinates of N stations determined by image processing*

Output: *optimized cleaning consumption*

Main Goal: *all points have maximum 2 selected paths(V=[1,1,2,2,…])*

Sub-Goal-1: *selected paths expose one closed loop*

Sub-Goal-2: *no intersecting paths in the loop*

```
1  Calculate distance(consumption) matrix using eq.9
2  while: not all open loops are connected (or closed loop.count>0)
       Group open loops and Create Lookup Table (Fig.15)
       n=open loop count
       Select n-1 minimum elements of sorted distances which are not selected
       Determine Visited Count vector(V)
       while: not (min(V)==1 and max(V)==2) (or all points are not blue or green)
           Define soft violations (yellow) and hard violations (red)
           if: hard violations count>0
               Tournament elimination among hard violations
           else:
               Tournament elimination among soft violations
           end
           Define SDM (Fig.14) and Tournament selection among uncolorod squares
           Uptade Lookup Table and V vector
       end
       Convert closed loops to open loop based on tournament elimination (Fig.15)
   end
3  Convert last one open loop to closed loop to present TSP-route
4  Apply Path-Intersection Optimizer (PIO)
```

2.3.2.2. Tournament selection-based lookup table refinement. In the RTVA, a Tournament Selection-based mechanism is employed to iteratively correct and refine the solution derived from the initial lookup table. The goal is to gradually evolve a set of valid and energy-efficient paths, ultimately forming a single open-loop tour that visits all contaminated areas with minimal cost.

The threshold parameters used for validation stage — including the degree constraint, the number of selections per stage, recursion depth, and path selection/elimination priorities — were determined empirically and logically grounded in routing principles. These thresholds were tuned to achieve stable convergence across multiple test configurations while minimizing unnecessary recursion cycles. For example, when applying the open-loop grouping concept (Section 2.3.2.3), an edge selection count of $n - 1$ was preferred over n to preserve this structure during the recursive validation process. However, in a valid TSP route, there are n selected paths, and each node has a degree of 2. If the edge selection count had been set to n, the algorithm would have faced difficulties in achieving Sub-goal-1, as premature loop closures would prevent proper open-loop grouping. The elimination priorities were optimized to balance the computational load according to the problem configuration. Overall, these parameters were chosen to ensure the fastest and most stable convergence during validation.

2.3.2.3. Open-loop initialization, grouping and path selection. Initially, all contaminated points are considered as individual open loops, and a new lookup table is created using only the vertex connections of these loops at all stages. The corresponding grouped paths are marked in green, indicating that they are fixed selections and will be excluded from subsequent deletion steps (Fig. 15).

From the remaining unmarked paths in the table, the algorithm selects $n-1$ minimum-cost connections—where n is the number of open loops at that stage. The total number of possible paths is $(n-1)/2$, and green-marked paths are excluded from the selection pool. In each iteration (as described in Pseudocode-1), the worst-performing path (red-marked as high-priority) is eliminated using tournament selection, and the most suitable alternative (one that satisfies all path rules) is added in its place. Candidate paths are filtered using a selectable distances matrix (Fig. 14), which includes only paths that do not violate the degree constraint.

2.3.2.4. Handling closed loops and ensuring global connectivity. As iterations proceed, closed loops may emerge in the solution, resembling outcomes of a Minimum Spanning Tree (MST) algorithm [59], where nodes are connected via shortest paths. However, while MSTs focus on connectivity, they do not ensure full traversal without repetition—a critical requirement in TSP contexts.

To resolve this, RTVA employs a recursive loop-breaking strategy:

1. Closed loops are systematically converted to open loops, again using tournament selection as shown in Fig. 15.
2. The algorithm identifies corner nodes of the remaining open loops.
3. A new lookup table and connection vector V are constructed, this time treating the open loops as higher-level nodes.

The tournament selection process is then re-applied recursively on this reduced structure, merging smaller open loops into larger ones until only a single open loop remains, representing the optimal or near-optimal TSP route.

This adaptive refinement process not only improves solution feasibility but also reduces computational complexity by **constraining the search space** to rule-compliant path combinations throughout optimization.

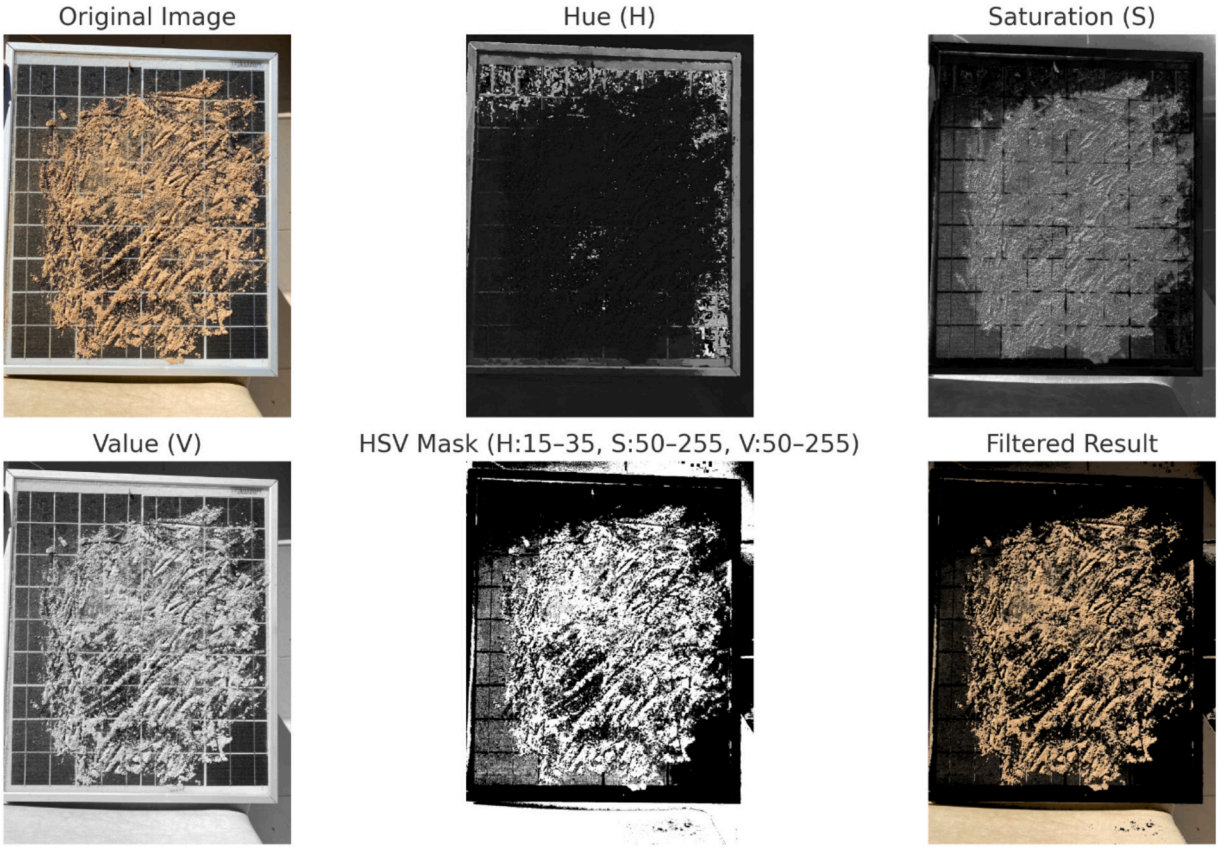

Fig. 11. HSV decomposition and filtering results for a dust-accumulated PV panel image.

2.3.3. Path-intersection optimizer (PIO)

The second phase of the RTVA focuses on refining the path topology by eliminating intersecting paths, which lead to inefficiencies in energy and time consumption. Intersections typically result in duplicated travel distances and suboptimal routing. Therefore, a Path-Intersection Optimizer (PIO) is introduced as a post-processing step to improve the route generated in the first phase.

After connecting the identified start and end stations to form a full route, the algorithm identifies path intersections by examining sequences of four points in the route. An intersection occurs when two non-consecutive edges cross each other, forming a structure that connects four stations but requires redundant traversal.

To resolve this, the algorithm:

- Identifies intersecting path pairs.
- Calculates the two possible alternative path pairs that could connect the same four stations without intersection.
- Chooses the one with lower total distance, similar in principle to the 2-Opt local search method [60].

However, unlike standard 2-Opt, which halts if no improvement is found, PIO continues searching. If the lower-cost pair (yellow paths in Fig. 16) results in an invalid closed loop, it is discarded in favor of the other. This ensures the continuity of a single closed-loop tour.

The algorithm checks for intersections using a clockwise ordering condition, which compares the geometric orientation of the segments. When an intersection is detected, the two intersecting edges are removed, and both alternative pairings are evaluated (Pseudocode 2).

Pseudocode 2. Path-Intersection Optimizer (PIO)

Pseudocode-2: Path-Intersection Optimizer (PIO)

Input: *route*

Output: *optimized route*

C_i**:** *Nodes in the route,* (C_i, C_j)*:Edges*

```
For i=1 to n-2:
    For j=1 to n-2 and (i-j) is positive number
        if: edges (Ci,Ci+1) and (Cj,Cj+1) intersect (clockwise ordering check)
            Replace intersections with (Ci, Cj) and (Ci+1, Cj+1)
        end
    end
end
```

2.3.4. Performance considerations

In TSP algorithms, convergence speed is as important as the accuracy of the solution. In comparisons between GA and Nearest Neighbor Algorithm (NNA), it is noted that although GA has slower computation speed, it is more effective than NNA in large-scale problems. On the other hand, NNA is superior in terms of speed and accuracy for smaller problems [61]. Various improvements have been proposed to reduce the greedy nature of the NNA [62–65]. While NNA selects the nearest path sequentially, it may miss better routes that could be encountered in subsequent steps. Greedy algorithms, by always selecting the shortest paths, are prone to getting trapped in local minimum. An overview of feature-level differences between existing algorithms is summarized in Table 3.

Although RTVA shares certain conceptual similarities with traditional graph-based and heuristic algorithms, its structure and objectives differ fundamentally. In the initial phase, RTVA rapidly selects $n - 1$ minimum-cost edges in a manner that resembles a MST; however, unlike Prim or Kruskal algorithms—which enforce connectivity and cycle constraints during edge selection but do not aim to construct a traversal route—RTVA performs these selections without preliminary validation. The algorithm then applies recursive table validation and sub-goal checks to refine these preliminary connections into a valid TSP-compliant path. In contrast, MST algorithms are primarily designed for connectivity optimization rather than traversal, as in network or transmission-line design problems. Prim's method resembles the NNA in its greedy behavior, evaluating grouped-to-ungrouped nodes to ensure connectivity without forming a visiting order, whereas Kruskal's approach sequentially sorts all possible edges while avoiding cycles, again failing to produce a valid traversal sequence.

The RTVA-PIO can also be compared with 2-Opt and Tabu Search. While 2-Opt removes intersections through pairwise edge swaps, it terminates once no immediate improvement is found, thus failing to guarantee a fully intersection-free route. Tabu Search, on the other hand, is a heuristic approach that explores neighboring solutions by prohibiting recently visited paths through a tabu list. Although this mechanism helps prevent premature convergence, it relies on numerous iterative trials and tabu-memory updates, resulting in high computational and memory overheads—making it unsuitable for embedded or real-time systems. In contrast, RTVA-PIO achieves deterministic intersection avoidance during recursive construction without requiring any memory structures or iterative neighborhood exploration.

Meta-heuristic algorithms such as GA, PSO, and ACO explore a vast number of route permutations to achieve near-optimal solutions, but this exhaustive search leads to high computational costs. RTVA mitigates this by pruning redundant regions of the search space through rule-based grouping and recursive validation, significantly accelerating convergence. In addition, unlike RTVA, reinforcement learning–based routing requires a prior training phase to develop an effective policy, which may limit its applicability in IP-based prior detection or hardware-constrained systems. This technique was not simulated in this study because its baseline conditions differ from other methods, as it requires simultaneous contamination detection and environmental parameter feedback.

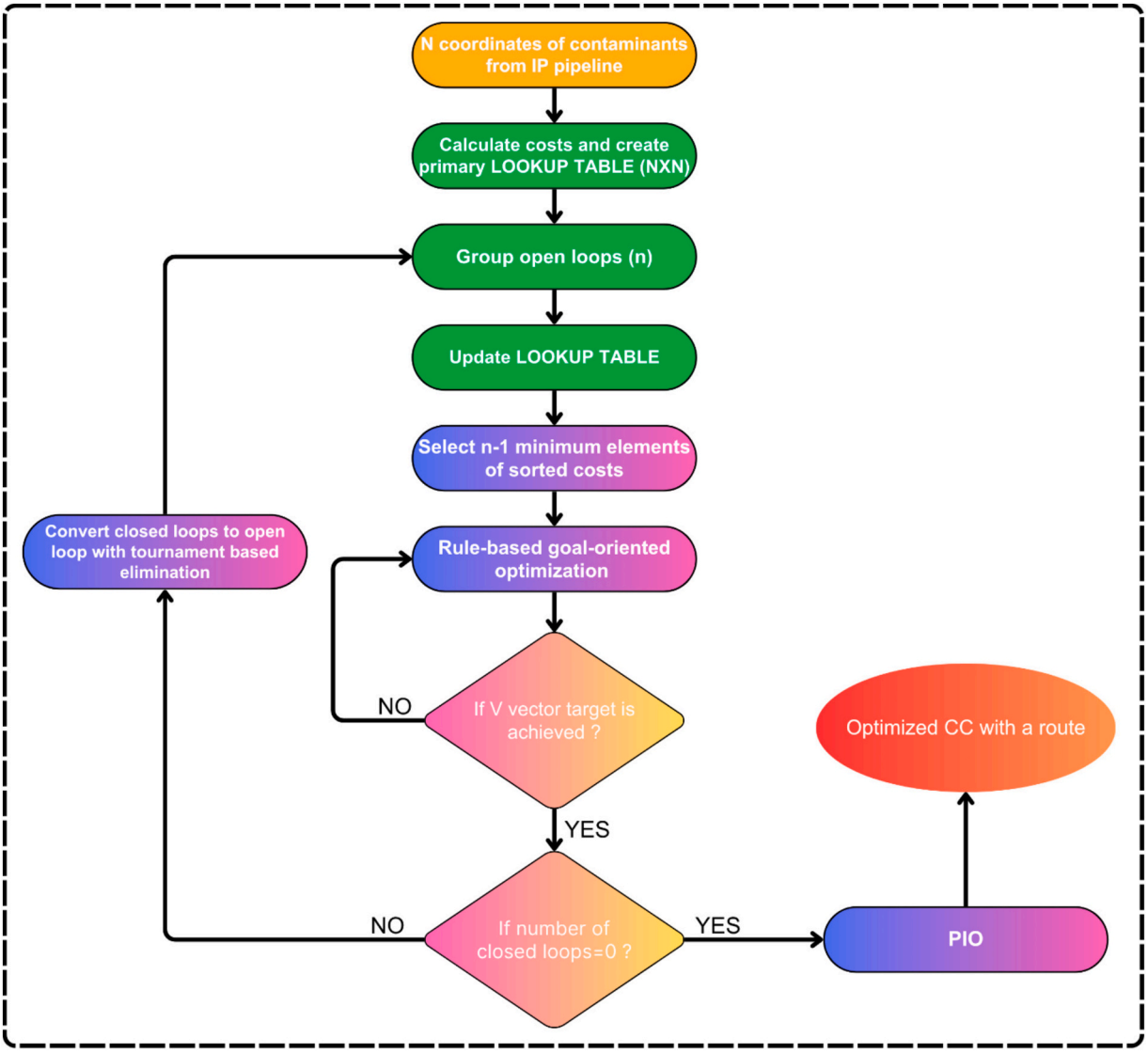

Fig. 12. Flow diagram of the proposed RTVA

Nevertheless, a practical limitation of RTVA is that, while rule applications are fast and deterministic, the PIO stage may exhibit variable execution times depending on the geometric configuration of the problem.

3. Experimental results and discussion

3.1.1. Image processing results

The proposed autonomous solar panel cleaning system, combined with image processing and the RTVA, was evaluated through a series of experiments conducted on real-world panel images. The image processing pipeline successfully identified contamination coordinates, with the results of each image processing step shown sequentially in Fig. 17.

Each processing step contributes to refining the input for the path optimization algorithm. The final binary contamination mask highlights localized soiling regions rather than full-surface accumulation, enabling targeted cleaning. The extracted contamination coordinates were transmitted from the processing computer (Python-OpenCV environment) to the Arduino controller, forming the real-time input for the route optimization and energy modeling subsystems.

3.2. Path optimization and energy consumption results

Following contamination detection, the system calculated energy consumption based on Eq. 9, which models the electrical power usage of each stepper motor during motion along the X and Y axes. The lookup table constructed from these energy costs was used as the base cost matrix for all routing algorithms.

Fig. 18 presents the complete RTVA routing process in four sequential stages. In Fig. 18a, the algorithm begins with greedy minimum-distance selections, forming the initial raw path configuration. Fig. 18b illustrates the point at which the main goal is achieved—each node satisfies the degree constraints (two connections per intermediate node and one per terminal node)—and in this particular case, sub-goal-1 (single closed-loop formation) is also met incidentally. Fig. 18c shows the initiation of sub-goal-2, where the PIO begins to correct intersecting or redundant paths. Finally, Fig. 18d depicts the stage at which all intersections are eliminated, resulting in a continuous and smooth traversal route that connects all contaminated regions efficiently.

This integrated visualization demonstrates how RTVA progressively transforms an initially greedy solution into a topologically valid, intersection-free path that minimizes both total travel distance and cleaning energy consumption. A comparative performance analysis with

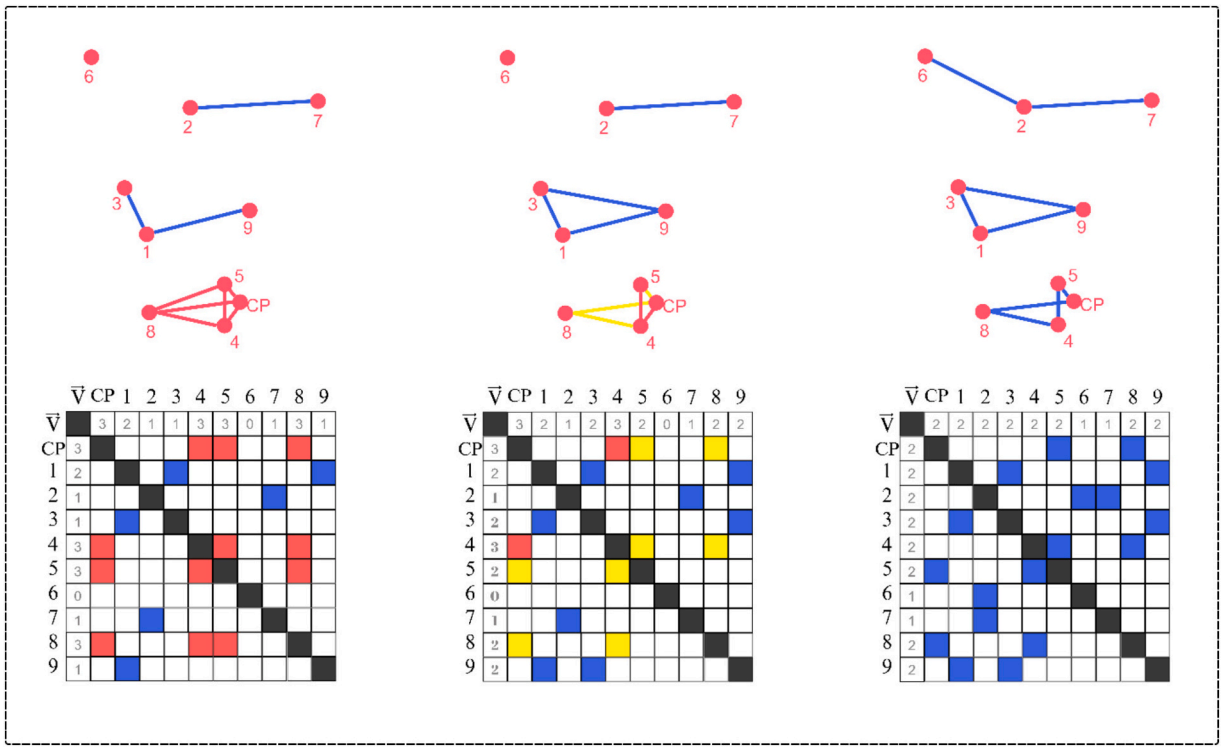

Fig. 13. RTVA lookup-table recursion (eliminating violating edges and selecting goal-oriented paths to achieve the main goal).

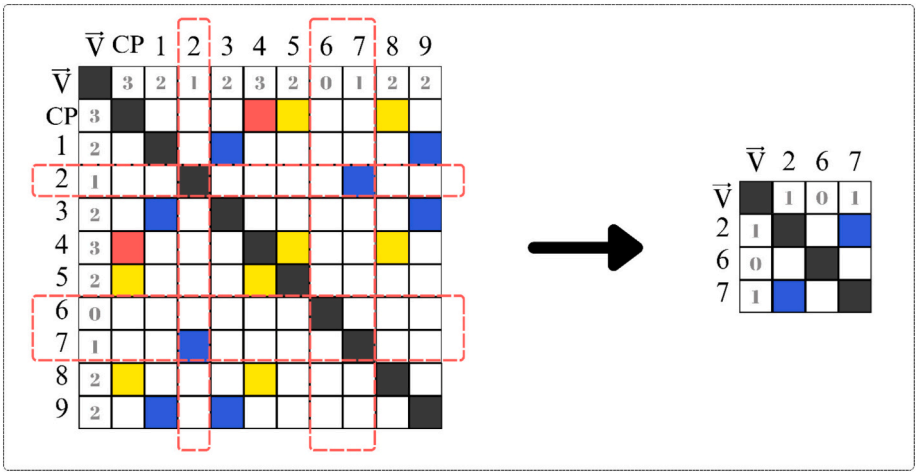

Fig. 14. Selectable distance matrix after rule-based filtering.

other classical TSP methods is presented in Table 4.

RTVA demonstrated the lowest cleaning energy consumption, achieving 19 % savings compared to the baseline and outperforming conventional greedy and local optimization methods. Notably, it achieved this improvement at an almost identical runtime to the 2-Opt algorithm, demonstrating that its recursive path validation and energy-aware route selection mechanism provides superior efficiency without compromising execution speed. This performance advantage stems from RTVA's ability to eliminate redundant motion before optimization rather than correcting it afterward.

The visual comparison in Fig. 19 further supports this observation, where suboptimal crossings and longer routes are visible in paths generated by (a) NNA and (b) NNA + 2-Opt, whereas the RTVA route remains smooth and intersection-free.

The measured energy consumptions—1605 J, 1542 J, 1490 J, and 1283 J for the IP-only, NNA, NNA + 2-Opt, and RTVA routes, respectively—showed only minor deviations from the calculated values. These small discrepancies can be attributed to voltage fluctuations in the grid-connected power supply during operation and idle waiting on soiled points.

3.3. Scalability and multi-scenario performance analysis

To evaluate scalability and robustness, the RTVA algorithm was tested under four synthetic contamination scenarios consisting of 20, 40, 100, and 250 nodes. Each dataset was optimized using six algorithms: RTVA, NNA, NNA + 2-Opt, GA, PSO, and ACO.

For geometric interpretability and visual comparison, the simulation cost was defined as the Euclidean distance between nodes, representing the energy consumption of the brush movement. Hence, the total route length directly corresponds to the total energy cost (*CC*).

Figs. 20–23 present the routing results for 20-, 40-, 100-, and 250-

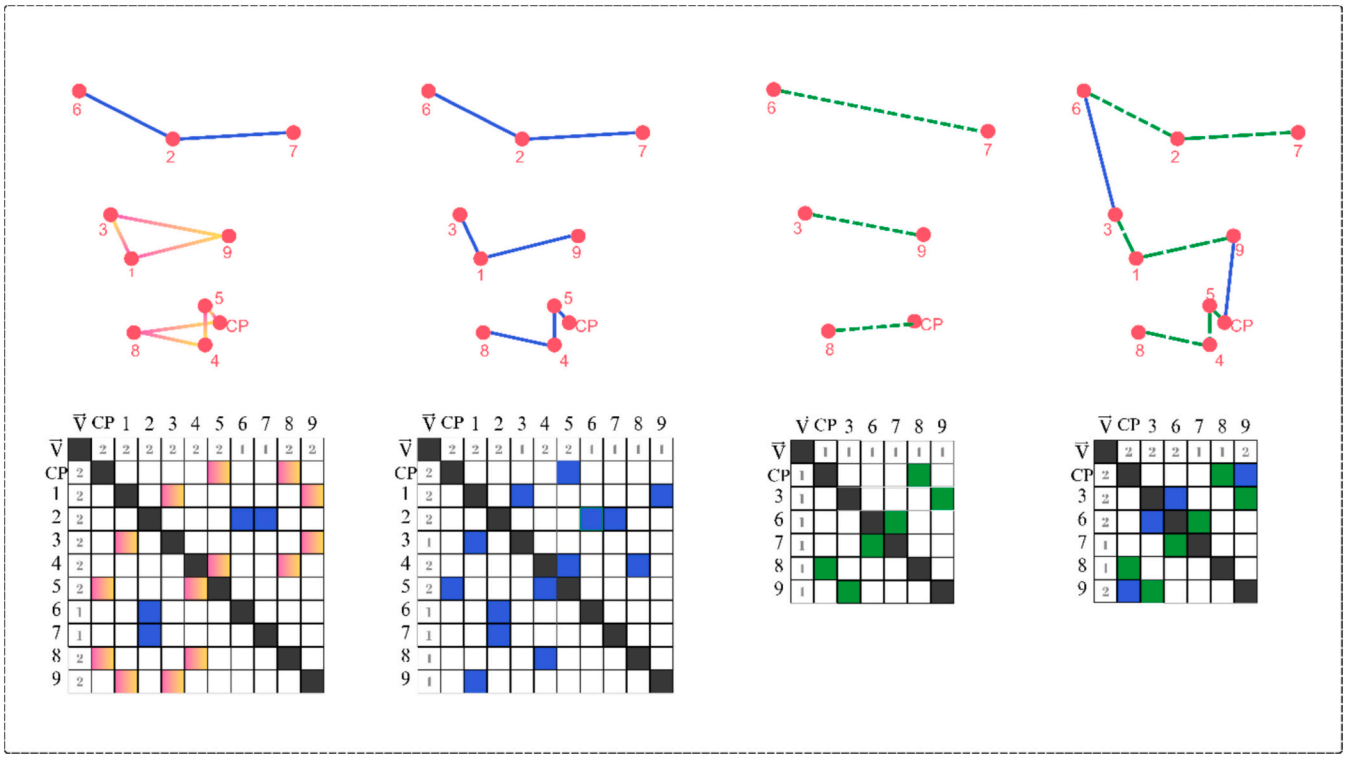

Fig. 15. Conversion of closed loops into open loops by narrowing the lookup table with internal open-loop connections.

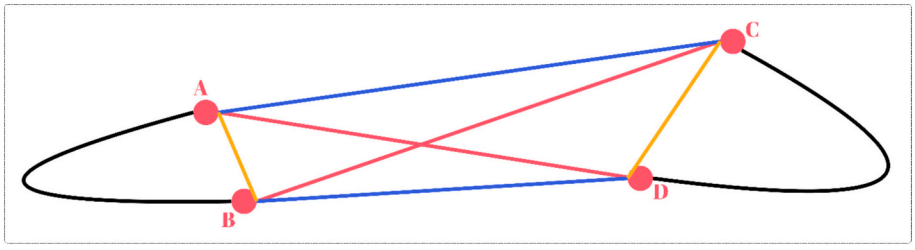

Fig. 16. Concept of the Path-Intersection Optimizer. (Black paths represent the selected routes of other stations, red paths indicate intersections, while yellow and blue paths represent alternative connections among the four stations). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

node scenarios, respectively. Each figure includes six subplots corresponding to NNA, NNA + 2-Opt, GA (first trial), RTVA (main-goal achieved), RTVA (sub-goal 1 achieved), and RTVA (sub-goal 2 achieved). GA results demonstrate the stochastic variability typical of meta-heuristic methods, while RTVA stages show progressive convergence from greedy initialization to fully validated, intersection-free paths.

Each figure presents routing results for increasing problem sizes, comparing RTVA with classical and meta-heuristic algorithms. RTVA maintains route smoothness and convergence stability across all scales. Simulation parameters were as follows: GA – population = 100,

Table 3
Benchmarking of existing optimization algorithms.

Algorithm	Main Focus	Intersection Prevention	Runtime	Similarity	Limitation
MST [59]	Global connectivity	Partial (but not in a TSP tour)	Low	Edge selection	Ensures global connectivity but not a route (TSP).
NNA [61]	Node order selection	✖	Low	Tournament selection	Dependent on initial path; may trap in local minima
2-Opt [60]	Local edge-swap optimization	Partial	Medium	PIO	Cannot prevent all intersections.
Tabu [54]	Iterative neighborhood search with tabu memory	Partial	High	Rules and tabu	High computational cost and memory usage; unsuitable for embedded systems.
Meta-heuristics [49,50,52]	Population-based node order optimization	✖	High	✖	Requires numerous iterations
Reinforcement-Learning based routing [71]	Experience-driven, adaptive path optimization	◆ (learned via reward shaping)	Medium–High	Policy-driven decisions	Requires prior training; sensitive to reward and exploration policy design.
RTVA	Rule-based goal-oriented Edge selection	◆	Medium-Low	◆	Execution time may vary with problem geometry with PIO

iterations = 300; PSO – population = 50, iterations = 300; ACO – population = 20, iterations = 100. These parameter values were selected through preliminary experimental tuning for large-node scenarios to ensure algorithmic stability and convergence, as lower population or iteration counts caused premature stagnation. Although smaller parameter sets may suffice for low-dimensional problems, larger configurations were required here to achieve consistent convergence across higher node counts. This setup provides a fair comparison among all methods while maintaining scalability and reproducibility. The corresponding runtime and cost performances are summarized in Table 5.

As summarized in Table 5, the runtime of RTVA exhibits an approximately quadratic growth ($O(n^2)$) with the number of nodes, consistent with its theoretical complexity described in Section 2.3.4. For comparison, NNA demonstrates similar $O(n^2)$ behavior, while NNA + 2-Opt shows $O(n^2 \cdot k)$ characteristics due to pairwise swap iterations. In contrast, meta-heuristic algorithms such as GA, PSO, and ACO follow cubic-to-quartic growth trends depending on their population and iteration structures. Although these algorithms maintain relatively stable runtime increases across node counts, their convergence to optimal or near-optimal routes becomes less reliable as problem size increases. This is mainly because, in large-node problems, the higher-iteration configurations used to ensure convergence also amplify stochastic variability, often yielding sub-optimal results within the tested runtime limits. In comparison, RTVA achieves faster and more consistent convergence than 2-Opt at higher node counts, confirming its suitability for embedded and real-time implementations where deterministic timing and computational efficiency are essential.

3.4. Discussion of results

RTVA consistently achieved the lowest average cost across all scenarios while maintaining near-linear runtime scaling. Its early pruning and grouping, with rule-based path validation significantly shortened convergence time, enabling efficient optimization even for the 250-node problem. Although NNA exhibited the fastest raw computation time due to its greedy nature, it frequently converged to sub-optimal routes, failing to reach global optima. The inclusion of the 2-Opt enhancement improved the initial NNA solution, yet RTVA's PIO achieved both lower cost and shorter runtime, effectively resolving all path crossings instead of partial correction.

Among meta-heuristic methods, runtime demands increased drastically with problem size—often by several orders of magnitude. While PSO showed relatively faster performance and a milder runtime growth rate than GA or ACO, its minimum achieved costs still exceeded those of RTVA. In large-scale scenarios, ACO exhibited the closest performance in route optimality but required excessive computation time (≈130 s for 250 nodes)—over 30 times slower than RTVA—and is therefore unsuitable for real-time or embedded robotic applications. Notably, RTVA executed nearly 1000 times faster than ACO in small-scale scenarios.

Overall, the RTVA algorithm outperformed all competing methods in terms of cost efficiency and computational scalability. In large-scale implementations, where motor energy consumption increases proportionally with travel distance, the energy-saving advantage of RTVA becomes even more pronounced. Its deterministic and rule-driven architecture eliminates stochastic variability, producing consistent results across repeated runs, unlike the probabilistic behavior of GA, PSO, and ACO.

3.5. Discussion on sustainability impacts

The proposed robotic cleaning system contributes to the long-term sustainability of PV maintenance by reducing both resource consumption and mechanical degradation. The system features a precisely actuated dual-axis cleaning module capable of targeted motion across the panel surface, enabling efficient operation under non-uniform soiling conditions. Unlike full-surface cleaning, this configuration allows energy-efficient and selective actuation, further enhanced by the integrated routing capability that optimizes the brush trajectory to minimize redundant motion.

In the small-scale prototype experiments, RTVA-based targeted cleaning achieved up to 20 % measured energy saving compared with the route determined solely by the IP pipeline. Considering that the robotic system completes a full-surface cleaning cycle in approximately 70 s, and assuming a 5 s cleaning duration per contaminated point, localized cleaning based only on IP outputs still achieves nearly 50 % savings in operational resources relative to full-surface operation. This reduction in unnecessary motion also improves water-use efficiency and minimizes mechanical wear. In rooftop installations, this not only improves operational efficiency but also reduces occupational safety risks associated with manual cleaning.

Although periodic verification is required to mitigate sensor drift or detection errors, the integration of IP modules with environmental decision-making sensors (e.g., irradiance, temperature, and humidity) enables cross-validation of measurements, thereby reducing recalibration frequency or manual supervision. This synergistic sensing architecture enhances both the system's autonomy and its long-term reliability under field conditions.

Beyond terrestrial applications, the RTVA architecture offers distinct advantages for drone-assisted or hybrid aerial cleaning systems, which have recently emerged as an alternative to ground-based robots. Spraying drones [66,67] or air-jet cleaning units [68] face strict constraints on payload capacity, flight duration, and energy efficiency. In such platforms, minimizing redundant travel paths directly translates into extended flight endurance and greater cleaning coverage. Because RTVA emphasizes energy-aware edge selection rather than node ordering, it is inherently well-suited for these lightweight aerial platforms, effectively reducing computational load, route energy, and operational time — as demonstrated in the scalability simulations.

Overall, the integration of image-based contamination detection with an energy-aware routing algorithm demonstrates a practical step toward sustainable, intelligent PV maintenance. The framework's modularity allows its adaptation not only to fixed installations but also to emerging mobile and aerial cleaning systems, providing a versatile foundation for future large-scale autonomous solar maintenance infrastructures.

3.6. Limitations

While the proposed RTVA-based framework demonstrates significant potential for autonomous PV cleaning, several simplifying assumptions and practical limitations should be considered when interpreting the results.

First, the cleaning efficiency of the rotating brush mechanism was assumed to remain constant throughout the experiments. In practical applications, this efficiency may fluctuate due to variations in surface roughness, humidity, or dust particle adhesion. For future aerial or hybrid mobile cleaning systems, this limitation could be mitigated by employing adaptive-pressure cleaning heads, or even vacuum-assisted modules [69] that dynamically adjust contact force according to contamination intensity. Although such mechanisms were not implemented in this study, the modular nature of the proposed architecture can readily accommodate them. As an additional extension, steam-assisted cleaning may provide further advantages by softening particulate adhesion before mechanical brushing, thereby reducing wear and improving efficiency—particularly for stubborn or adhesive contamination layers.

Second, the real-time decision-making framework relies on predictive models that estimate future power losses based on irradiance and temperature, while the overall energy gain of the framework is beyond the present scope, as the study primarily focuses on minimizing energy consumption. In this decision-making process, the panel temperature is assumed to vary only with solar irradiance and ambient temperature

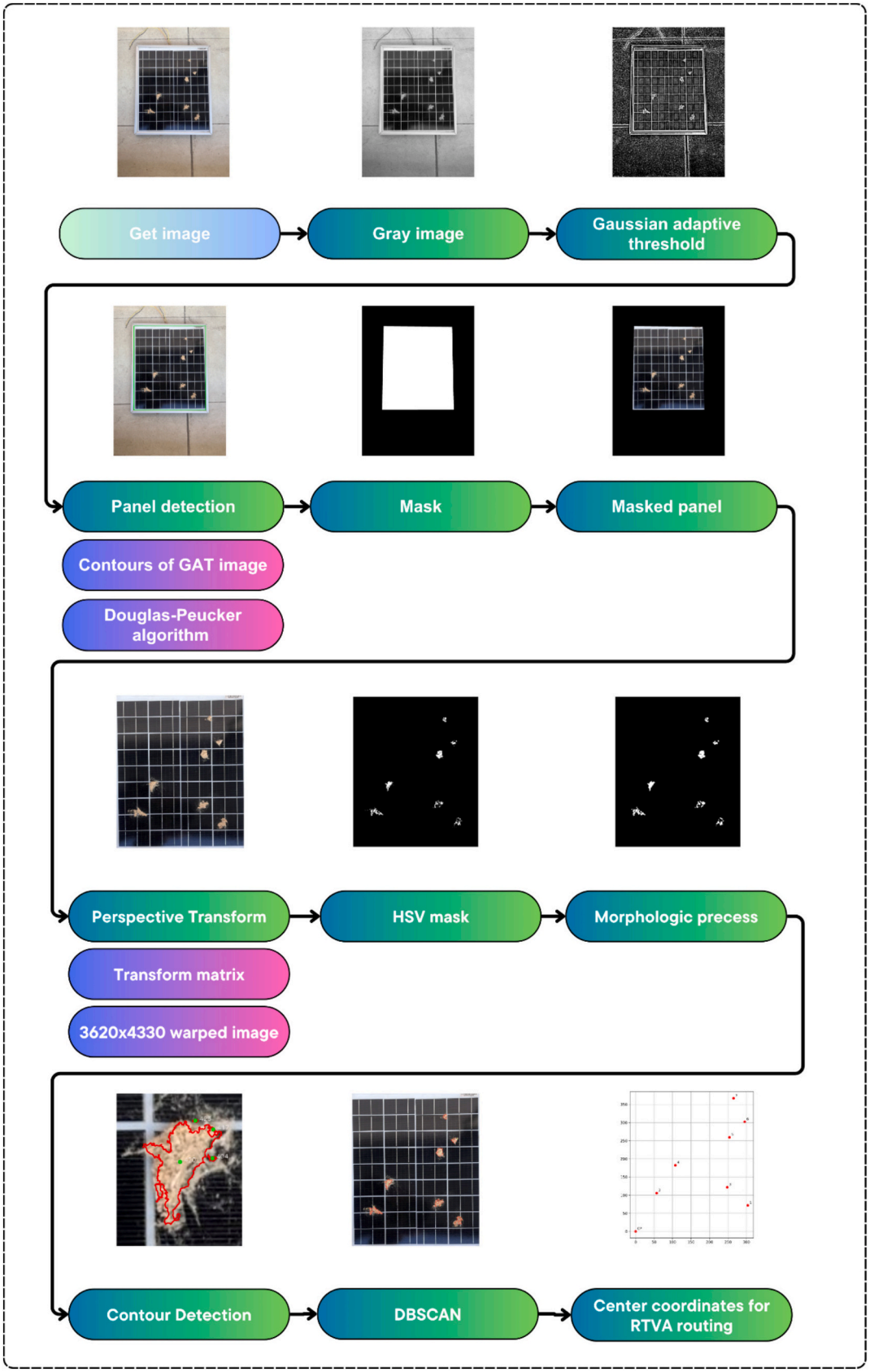

Fig. 17. Sequential image processing stages for contamination detection.

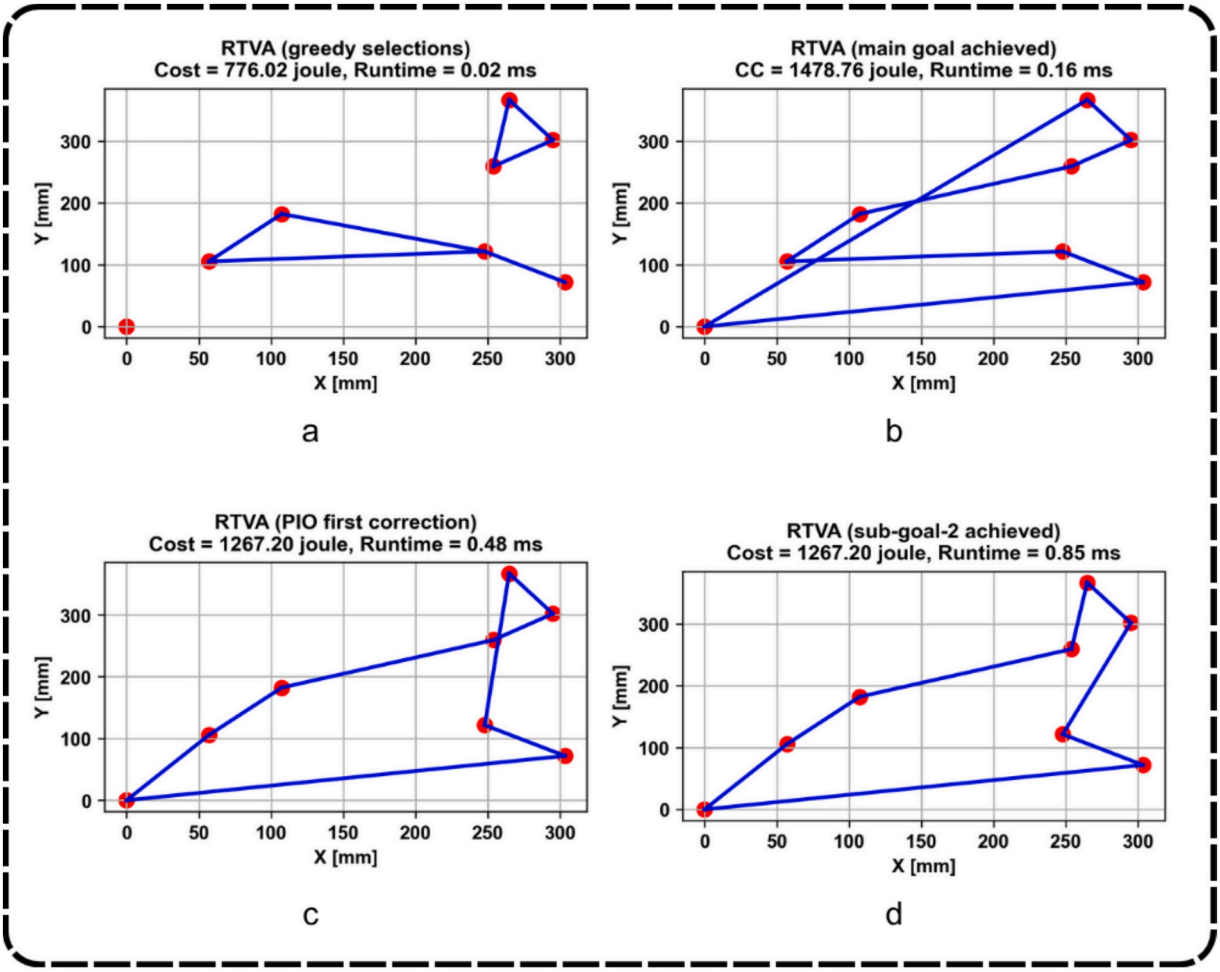

Fig. 18. RTVA route formation and optimization stages: (a) Greedy selections; (b) Main goal achieved (sub-goal 1 also achieved); (c) PIO first correction; (d) Sub-goal 2 achieved.

Table 4
Comparative performance of route optimization algorithms.

Algorithm	Cleaning Consumption [J]	Recovery Percentage	Runtime [ms]
Based on IP-only route	1561	–	
NNA	1521	%3	0.04
NNA + 2-Opt	1478	%6	0.76
RTVA	1267	%19	0.85

according to the panel datasheet. However, previous studies have demonstrated that this estimation can be improved by incorporating correlation models that also account for humidity and wind speed effects [28]. Moreover, non-uniform contamination across the panel configuration can cause local Maximum Power Point (MPP) variations within contamination-induced shaded regions [70]. In the current framework, however, the detected contaminated areas are assumed to affect output power uniformly as a function of irradiance, meteorological parameters, and detected instantaneous efficiency. For more precise adjustment, the system could integrate IP outputs to spatially map the transmittance distribution of non-uniform soiling across the panel configuration. By applying equivalent-circuit modeling to these localized transmittance maps, future implementations could achieve higher prediction accuracy and more realistic power-loss estimation under heterogeneous contamination conditions.

Finally, the IP pipeline was implemented and validated at a small scale using fixed-camera inputs to extract panel boundaries and contamination contours. Within this framework, the HSV-based image segmentation approach provided computationally efficient results; however, its reliability may decrease under extreme lighting variations or non-uniform surface reflections. In addition, the HSV-based segmentation has not yet been tested across different contamination types or panel materials. The threshold parameters used in this study were tuned specifically for the glass surface and coating characteristics of the tested module; therefore, further validation on alternative panel configurations, such as polycrystalline or textured surfaces, is required to ensure robustness and general applicability. Although controlled camera calibration and empirical threshold tuning partially mitigated these limitations in the current prototype, greater robustness and scalability can be achieved through dynamic thresholding and lightweight ML-based segmentation. For larger-scale or outdoor deployments, this stage could be enhanced further by employing DL-based detection frameworks that can be embedded into edge devices or onboard processors, which enable more robust and adaptive panel identification under diverse environmental conditions.

Despite these limitations, the experimental and simulated results validate RTVA's core advantages—fast convergence, energy-aware optimization, and adaptability. These constraints primarily stem from system-scale and environmental variability rather than algorithmic deficiencies, underscoring RTVA's potential as a foundational component

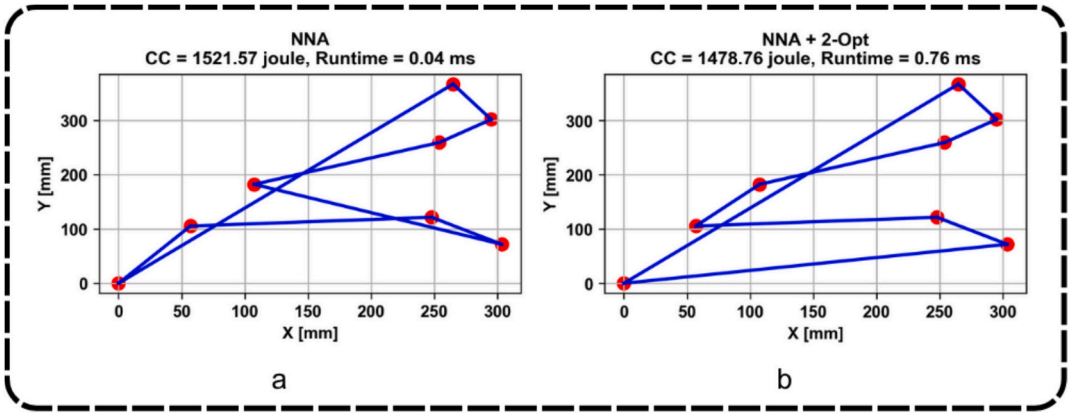

Fig. 19. Visual comparison of route efficiency a) NNA and b) NNA + 2-Opt.

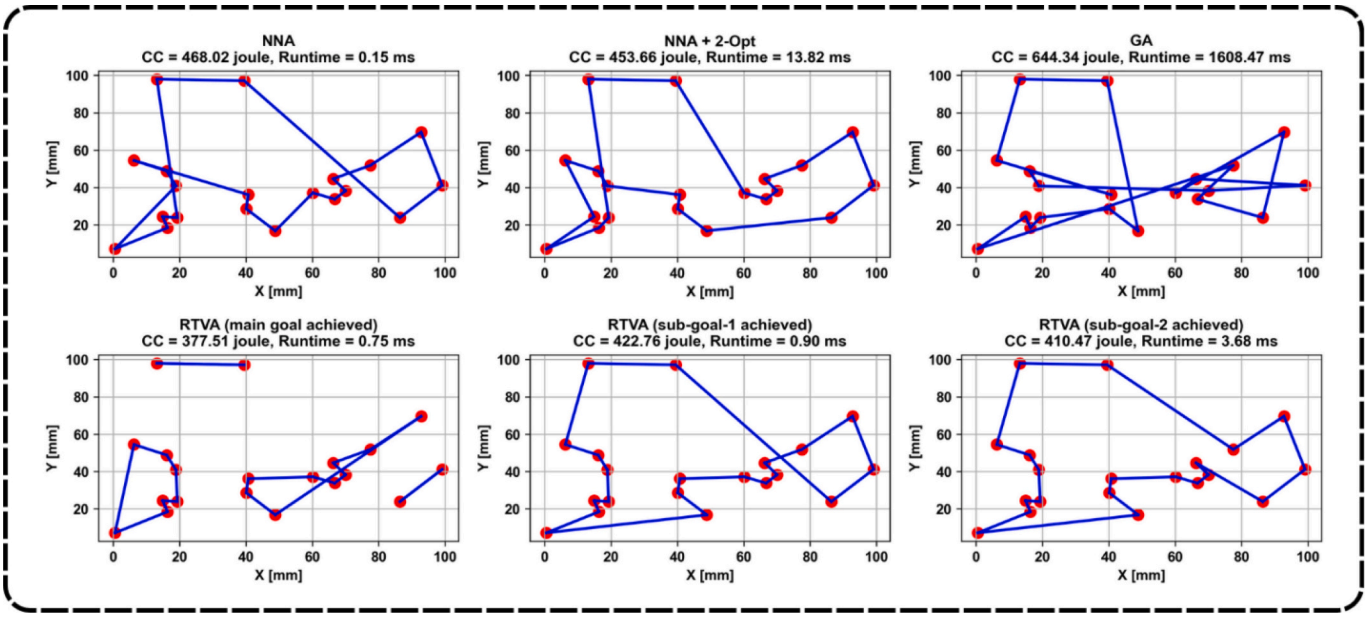

Fig. 20. Optimized routes for the 20-node scenario (NNA, NNA + 2-Opt, GA (first trial), RTVA (main-goal achieved), RTVA (sub-goal 1 achieved), RTVA (sub-goal 2 achieved) sequentially).

for next-generation autonomous and aerial solar cleaning systems.

4. Conclusion and recommendation

This study developed a fully autonomous solar panel cleaning system integrating a two-axis ball-screw-driven cleaning mechanism with a vision-based contamination detection and an energy-aware decision-making framework. Unlike conventional systems that clean at fixed intervals or cover the entire panel surface, the proposed design performs targeted cleaning only on detected soiled areas, minimizing unnecessary motion, water usage, and energy consumption.

Experimental validation demonstrated that the proposed Recursive Tabular Validation Algorithm (RTVA) achieved an average 19 % reduction in energy consumption and a comparable decrease in cleaning time, owing to the mechanical efficiency of the dual-axis brushing system. Large-scale simulations (20–250 nodes) further confirmed RTVA's scalability: while metaheuristic algorithms (GA, PSO, and ACO) exhibited high computation load, RTVA maintained low runtime with near-optimal route performance, achieving up to 21 % and 11 % lower total route cost compared to NNA and NNA + 2-Opt, respectively. These results confirm that RTVA's path-centric design effectively balances computational efficiency and route optimality, making it well suited for embedded and real-time robotic cleaning applications.

The significance of this work lies in its contribution to sustainable PV maintenance. By combining lightweight image processing with energy-aware routing, the proposed framework reduces water consumption, minimizes mechanical wear, and lowers maintenance labor. Its modular architecture allows seamless adaptation to various system scales—from rooftop installations to utility-scale solar farms, where payload and energy constraints are critical.

5. Recommendations

For practical deployment, several directions are recommended to extend the capabilities of the proposed framework. Within the RTVA structure, recursive refinements can be incorporated into the grouping stages, allowing each cluster to perform internal validation and optimization cycles. However, attention should be given to the potential increase in computational load introduced by these additional recursive processes, requiring an appropriate trade-off between accuracy and runtime. Furthermore, tournament-based selection steps across all stages could be enhanced through probabilistic or statistical evaluation

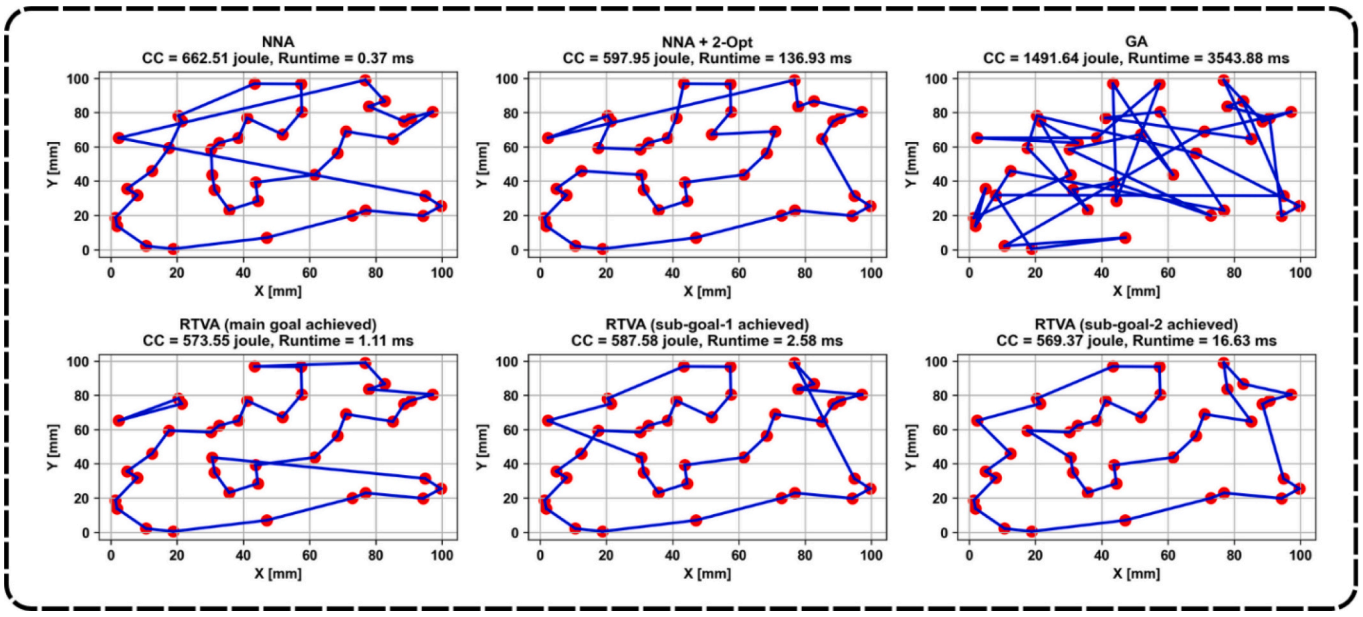

Fig. 21. Optimized routes for the 40-node scenario (NNA, NNA + 2-Opt, GA (first trial), RTVA (main-goal achieved), RTVA (sub-goal 1 achieved), RTVA (sub-goal 2 achieved) sequentially).

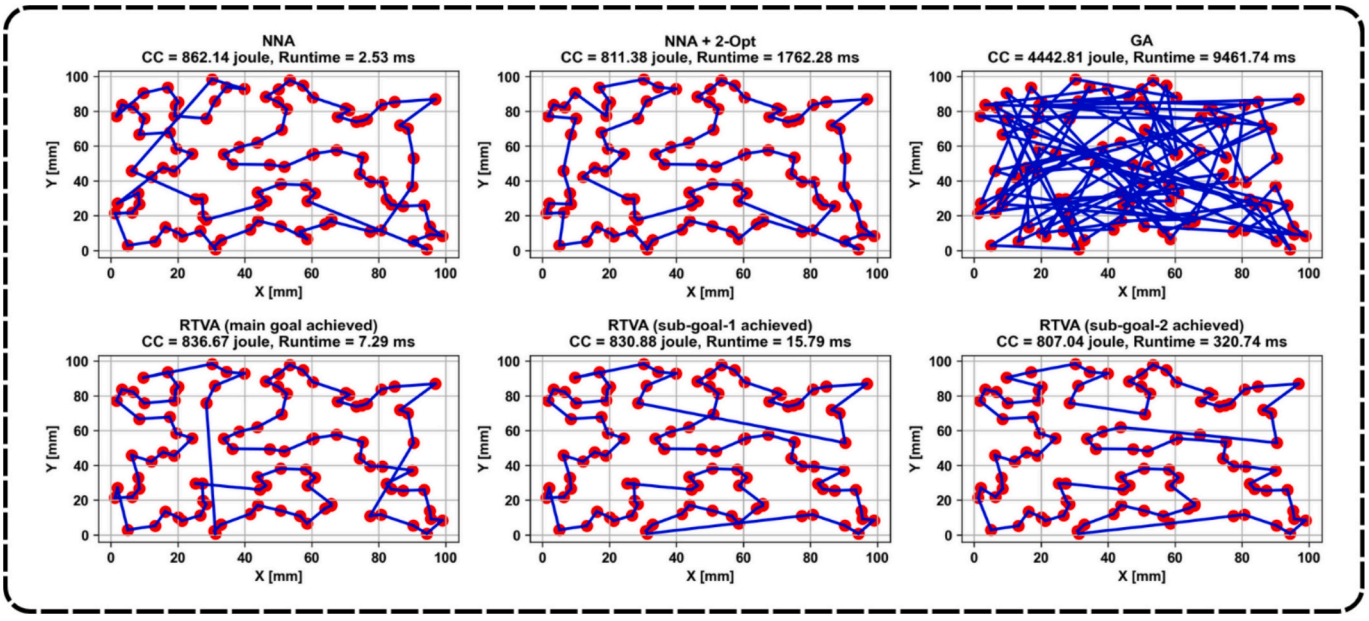

Fig. 22. Optimized routes for the 100-node scenario (NNA, NNA + 2-Opt, GA (first trial), RTVA (main-goal achieved), RTVA (sub-goal 1 achieved), RTVA (sub-goal 2 achieved) sequentially).

criteria, enabling more adaptive and robust decision-making during path elimination and edge validation.

Future implementations may also replace the current fixed-camera configuration with drone-based imaging and cleaning systems. Such aerial platforms can dynamically capture contamination data across large-scale PV fields and perform localized cleaning actions with higher coverage efficiency. The RTVA algorithm can optimize both flight and cleaning paths according to non-uniform soiling patterns, thereby reducing energy and water consumption, improving sustainability, and enhancing scalability for utility-scale solar farms. Beyond the simulation-based scalability analysis, future studies will prioritize field-scale validation of the proposed framework under real environmental and mechanical conditions. Conducting large-area experiments in utility-scale PV farms will enable assessment of RTVA's robustness against external disturbances such as uneven terrain, and long-term operational wear. These field experiments are expected to bridge the gap between laboratory-scale demonstrations and full-scale industrial implementation, providing empirical confirmation of RTVA's reliability and adaptability under real-world conditions. Moreover, for future multi-agent or drone-based cleaning architectures, the inherent grouping procedure of RTVA can facilitate task partitioning among agents, while synchronization of multiple RTVA instances and mitigation of communication delays will be critical for maintaining global route efficiency and preventing task overlap. To this end, future scaling efforts should prioritize cloud-connected architectures that enable coordinated decision-making and integration with predictive soiling-loss

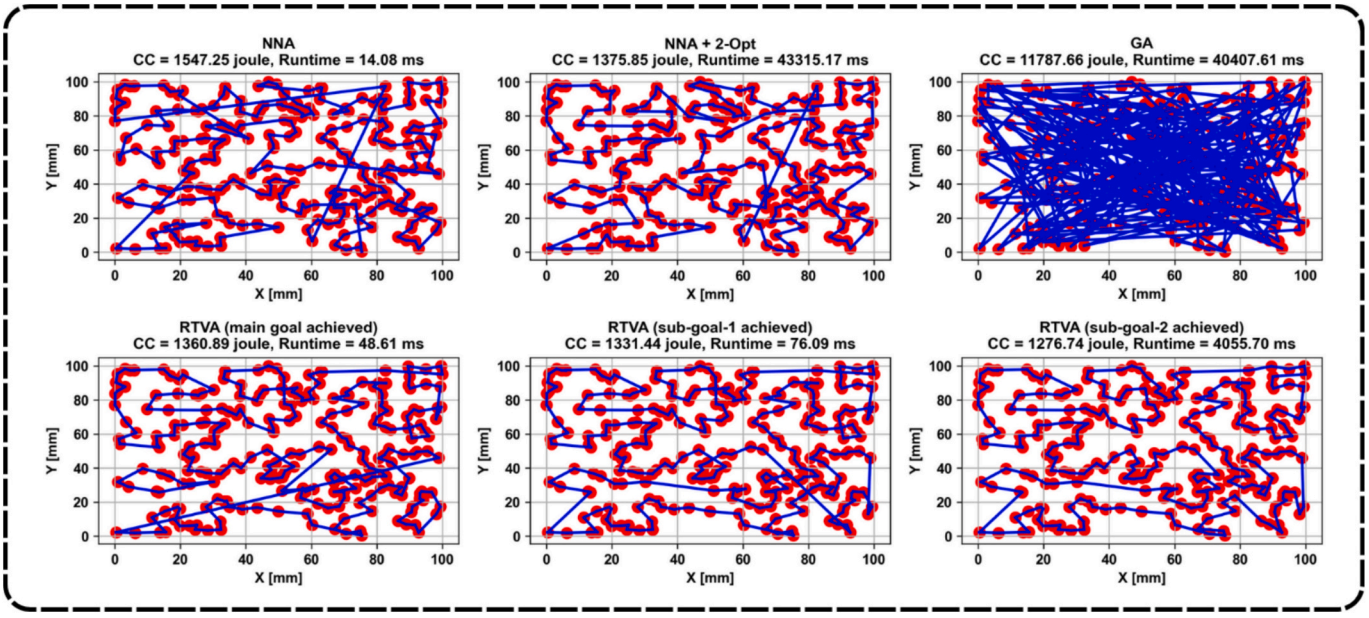

Fig. 23. Optimized routes for the 250-node scenario (NNA, NNA + 2-Opt, GA (first trial), RTVA (main-goal achieved), RTVA (sub-goal 1 achieved), RTVA (sub-goal 2 achieved) sequentially).

Table 5
Comparative performance of RTVA under different node-scale scenarios.

Algorithm	Nodes	Trials	Runtime [ms]	Route Cost [units]		
			(Mean)	Min.	Max.	Mean
	20	1	0.15	–	–	468.02
	40	1	0.37	–	–	662.51
NNA	100	1	2.53	–	–	862.14
	250	1	14.08	–	–	1547.25
	20	1	13.82	–	–	453.66
	40	1	136.93	–	–	597.95
NNA + 2Opt	100	1	1762.28	–	–	811.38
	250	1	43,315.17	–	–	1375.85
	20	10	3264.77	400.85	401.74	**401.74**
	40	10	7736.28	545.59	569.32	**559.11**
ACO	100	10	29,021	799.87	815.89	808.33
	250	10	129,853	1332.06	1380.15	1357.21
	20	10	319.71	593.77	711.54	670.34
	40	10	445.60	1512.72	1622.88	1571.70
PSO	100	10	990.56	4472.59	4650.49	4583.75
	250	10	2350.65	11,702.3	12,258.23	12,095.68
	20	10	2060.50	568.66	644.35	611.40
	40	10	3734.32	1435.92	1513.22	1467.36
GA	100	10	11,466	4220.08	4401.46	4320.62
	250	10	50,619	11,440.7	11,806.06	11,692.27
	20	1	3.68	–	–	410.47
	40	1	16.63	–	–	569.37
RTVA	100	1	320.74	–	–	**807.04**
	250	1	4055.70	–	–	**1276.74**

analytics across distributed PV farms.

These recommendations aim to translate the algorithm's laboratory success into field-ready, scalable, and sustainable PV cleaning solutions, providing a pathway toward next-generation intelligent and environmentally efficient solar maintenance systems.

CRediT authorship contribution statement

Emrah Üstündağ: Writing – original draft, Visualization, Software, Methodology, Data curation. **Fatih Onur Hocaoğlu:** Writing – review & editing, Validation, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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